Percolation and $\theta(p_c) = 0$

Abstract

These notes prove that the percolation probability vanishes at the critical probability on $\mathbb{Z}^d$ in every dimension, for bond percolation and for site percolation alike. Both follow from one inequality about a finite family of independent elements, each joining an arbitrary set of vertices, in which no lattice appears. Harris and Kesten settled the plane and the lace expansion settled dimension eleven and above; the argument here uses neither duality nor the lace expansion and reaches every dimension at once. The finite inequality is due to Anthropic, who proved it for bond percolation; running it in the general model is ours. The prerequisite is a first graduate course in probability. The whole argument, from the correlation inequalities to the two lattice statements, has been checked in Lean, and the final theorem is accepted with no hypothesis beyond the dimension and with no axiom beyond the three that Lean itself provides.

The mathematics in these notes was produced and audited by OpenAI Codex models. The exposition is by Claude Opus 5, an Anthropic model running in Claude Code. Ahmed Bou-Rabee prompted and directed the work throughout, set the plan, fixed the standards, and supplied the corrections.

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1 Bernoulli Percolation on the Hypercubic Lattice

Broadbent and Hammersley [7] introduced percolation in 1957 to model a fluid spreading through a porous stone, and posed the existence of the phase transition as the central question. Declare each edge of the lattice $\mathbb{Z}^d$ open with probability p and closed otherwise, independently across edges, and ask whether fluid entering at the origin can travel arbitrarily far. Write $\theta(p)$ for the probability that it can, that is, for the probability that the origin lies in an infinite connected component of open edges. As p increases more edges are open, so θ increases; it vanishes when p is small and it is positive when p is close to 1. There is therefore a threshold parameter p_c separating the two behaviors, and what happens exactly at that threshold is the subject of these notes.

This section builds the model from the ground up and proves two things about it. The first is that the threshold is a genuine interior point: for every $d \geq 2$ we have $0 < p_c < 1$, so there really are two phases to separate. The second is that θ is continuous at every parameter other than p_c. The question of whether θ is continuous therefore collapses to the single number $\theta(p_c)$, which is either 0 or the size of a jump. That is why the problem is usually posed as the equation $\theta(p_c) = 0$.

The order is as follows. Section 1.1 says why the proofs in these notes are written for a model more general than percolation on a graph, and what a reader interested only in bond percolation should do about it. Section 1.2 constructs the probability space and the class of events used throughout. Section 1.3 introduces open clusters and the percolation probability. Section 1.4 proves that the threshold is an interior point, and Section 1.5 proves continuity away from it. Section 1.6 records what is known about $\theta(p_c)$ dimension by dimension, which is what the rest of these notes is organized to settle.

The question has been open in the middle dimensions for a long time. Harris and then Kesten settled the plane by 1980, and the lace expansion settled all sufficiently high dimensions, most recently down to eleven, but the dimensions between three and ten resisted every method. The obstruction is easy to state: the planar arguments use duality, which has no analogue above two dimensions, and the high-dimensional arguments use the fact that two independent random walks in many dimensions rarely meet, which fails when there is not much room. What eventually worked was neither, and it does not look at the lattice at all until the very last step.

We fix three names. The number p is the **parameter**, the function θ is the **percolation probability**, and the threshold p_c is the **critical probability**.

Throughout this section $d \geq 2$ is an integer and p is a number in $[0, 1]$.

1.1 One inequality for two models

The theorem these notes prove is about bond percolation, and every step of the proof below can be carried out for bond percolation alone. We do not do that, and the reason is site percolation.

In **site percolation** the vertices, not the edges, carry the randomness: each vertex of $\mathbb{Z}^d$ is open with probability p and closed otherwise, independently, and two open vertices are joined when they are neighbors in the lattice. The cluster of the origin is the set of open vertices reachable from it through open vertices, the percolation probability and the critical probability are defined from it exactly as before, and the same question is asked. Site percolation is not a special case of bond percolation. Gladkov and Zimin [18] proved that no bond model reproduces a site model even approximately, and their obstruction already appears at a single vertex of degree three. A proof written for bond percolation therefore says nothing about site percolation, and the site statement would have to be proved again from the beginning.

It does not have to be proved again. Nothing in the proof below uses the fact that an edge has two endpoints. What the proof uses is three properties: the states of the edges are independent, opening an edge creates connections and destroys none, and conditioning on one cluster leaves the configuration away from that cluster alone. All three continue to hold when an edge is allowed to meet any number of vertices at once.

So Sections 2 to 5 are carried out in that model from the start: a finite vertex set, and finitely many elements, each meeting an arbitrary set of vertices and each open or closed independently of the others. Bond percolation is the case in which every element meets exactly two vertices. Site percolation is the case obtained by putting one element at each vertex and one extra vertex on each edge, as Section 6.1 explains. The single finite inequality proved in Section 5 therefore yields the lattice statement for both models, and Section 6 reads the two off.

A reader interested only in bond percolation loses almost nothing by reading "edge" for "hyperedge" and "the two endpoints of the edge" for "incidence set" everywhere. The list below says where that reading needs a second thought, and it points forward to the places concerned.

1. In Definition 2.7 the substitution is exactly the one just described, and the model becomes bond percolation on a finite graph.

2. The set $\mathcal{K}_S$ of Definition 2.8 becomes the set of open edges meeting the cluster. It cannot be replaced by the set of vertices of the cluster, and that is not an artifact of the general model: the example proving it, Proposition 2.9, is a path on three vertices.

3. The trace of (5) becomes the set of endpoints outside Z of the open edges meeting Z. The second inclusion of Lemma 2.10 can be strict here too, as soon as Z has two vertices with a common neighbor, so the bookkeeping it forces is not an artifact of the general model either.

4. Sections 3 to 5 then read word for word as statements about finite graphs. They are the bulk of these notes.

5. Of Section 6, only the first sentence of Example 6.1 and the bond half of Section 6.2 are used. The encoding of site percolation and the discussion of it may be skipped.

1.2 Configurations, events, and the product measure

We construct the probability space carefully, because every later argument conditions on events and compares measures, and those operations need a fixed underlying space.

Regard $\mathbb{Z}^d$ as a graph: its vertices are the points of $\mathbb{Z}^d$, and two vertices are joined by an edge exactly when they are at Euclidean distance 1. Write $\mathbb{E}_d$ for the set of edges. This edge set is countably infinite, and each vertex lies in exactly $2d$ edges.

Definition 1.1 (Configuration) A **configuration** is a function $\omega : \mathbb{E}_d \to \{0, 1\}$. An edge e is **open** in ω when $\omega(e) = 1$, and **closed** when $\omega(e) = 0$. The set of all configurations is $\Omega_d = \{0, 1\}^{\mathbb{E}_d}$.

An event is **determined by** a set S of edges when membership in it depends on ω only through the values $\omega(e)$ for $e \in S$.

Definition 1.2 (Cylinder Event) A **cylinder event** is a subset of Ω_d determined by some finite set of edges. The **product σ-algebra** $\mathcal{F}_d$ is the smallest σ-algebra containing every cylinder event.

Equivalently, a cylinder event is a finite union of sets of the form $\{\omega : \omega(e_i) = a_i \text{ for } 1 \leq i \leq n\}$, since there are only finitely many assignments to a finite set of edges. The events we care about, such as the event that the origin lies in an infinite open cluster, are not cylinder events.

Definition 1.3 (Bernoulli Bond Percolation) **Bernoulli bond percolation** with parameter p on $\mathbb{Z}^d$ is the probability measure $\mathbb{P}_p$ on $(\Omega_d, \mathcal{F}_d)$ under which the coordinates $\omega(e)$ for $e \in \mathbb{E}_d$ are independent and $\mathbb{P}_p(\omega(e) = 1) = p$ for every edge e.

Existence and uniqueness of $\mathbb{P}_p$ is the Kolmogorov extension theorem applied to a product of countably many copies of the Bernoulli measure on $\{0, 1\}$, so we take the measure as given. Write $\mathbb{E}_p$ for the corresponding expectation.

One structural feature of Ω_d drives nearly every argument in these notes. Order configurations pointwise: write $\omega \leq \omega'$ when $\omega(e) \leq \omega'(e)$ for every edge e, that is, when every edge open in ω is also open in ω'.

Definition 1.4 (Increasing Event) An event $A \in \mathcal{F}_d$ is **increasing** when $\omega \in A$ and $\omega \leq \omega'$ together imply $\omega' \in A$. The event A is **decreasing** when its complement is increasing.

Opening more edges can create connections but never destroy them, so every event asserting the existence of an open path is increasing.

The most useful device for comparing two parameters is a coupling that realizes all of them at once on a single probability space.

Definition 1.5 (Standard Coupling) Let $(U_e)_{e \in \mathbb{E}_d}$ be independent random variables, each uniform on $[0, 1]$, defined on a probability space with measure $\mathbb{P}$. For $p \in [0, 1]$ define a configuration ω_p by setting $\omega_p(e) = 1$ exactly when $U_e \le p$. The family $(\omega_p)_{p\in[0,1]}$ is the **standard coupling**.

Lemma 1.6. (The Standard Coupling Has the Right Laws and Is Monotone) *Let $(\omega_p)_{p\in[0,1]}$ be the standard coupling. Then ω_p has law $\mathbb{P}_p$ for every $p \in [0,1]$, and $\omega_p \le \omega_q$ pointwise whenever $p \le q$.*

Proof. Fix p. The variables U_e are independent, so the events $\{U_e \le p\}$ are independent, and each has probability p because U_e is uniform on $[0,1]$. Hence the coordinates of ω_p are independent with $\mathbb{P}(\omega_p(e) = 1) = p$, which is the defining property of $\mathbb{P}_p$ in Definition 1.3.

For the monotonicity, let $p \le q$ and let e be an edge with $\omega_p(e) = 1$. Then $U_e \le p \le q$, so $\omega_q(e) = 1$. Since e was an arbitrary edge, $\omega_p \le \omega_q$. □

A consequence of Lemma 1.6 is that $\mathbb{P}_p(A) \le \mathbb{P}_q(A)$ for every increasing event A and all $p \le q$: realize both parameters in the standard coupling and note that $\omega_p \in A$ forces $\omega_q \in A$. We use this comparison from now on without further comment.

1.3 Open clusters and the percolation probability

An **open path** is a finite sequence of vertices $x_0, x_1, \ldots, x_n$ in which consecutive vertices are joined by an edge and every one of those edges is open. Two vertices x and y are **connected**, written $x \leftrightarrow y$, when some open path starts at x and ends at y; every vertex is connected to itself by the path of length 0. Connection is reflexive, symmetric, and transitive, so it partitions the vertex set.

Definition 1.7 (Open Cluster) Let x be a vertex of $\mathbb{Z}^d$. The **open cluster** of x is the set $\mathcal{C}_x = \{\, y \in \mathbb{Z}^d : x \leftrightarrow y \,\}$ of vertices connected to x.

Definition 1.8 (Percolation Probability) The **percolation probability** is

$$\theta(p) := \mathbb{P}_p\big(|\mathcal{C}_{\mathbf{0}}| = \infty\big)\,,$$

the probability that the open cluster of the origin is infinite.

For each $n \ge 1$ let A_n denote the event that the origin is connected to some vertex at Euclidean distance at least n from it. Each A_n is determined by the finitely many edges meeting the ball of radius n, so each A_n is a cylinder event and in particular lies in $\mathcal{F}_d$. The events A_n decrease in n, and their intersection is the event that $\mathcal{C}_{\mathbf{0}}$ is infinite, since a cluster is infinite exactly when it contains vertices at arbitrarily large distance. Therefore

$$\theta(p) = \mathbb{P}_p\Big(\bigcap_{n\ge1} A_n\Big) = \lim_{n\to\infty} \mathbb{P}_p(A_n) = \inf_{n\ge1} \mathbb{P}_p(A_n)\,, \tag{1}$$

where the second equality is continuity of the measure along a decreasing sequence of events and the third holds because a nonincreasing sequence of numbers converges to its infimum. Identity (1) is used twice below, once for monotonicity and once for continuity.

Lemma 1.9. (The Percolation Probability Is Nondecreasing) *The function θ is nondecreasing on $[0,1]$, and $\theta(0) = 0$.*

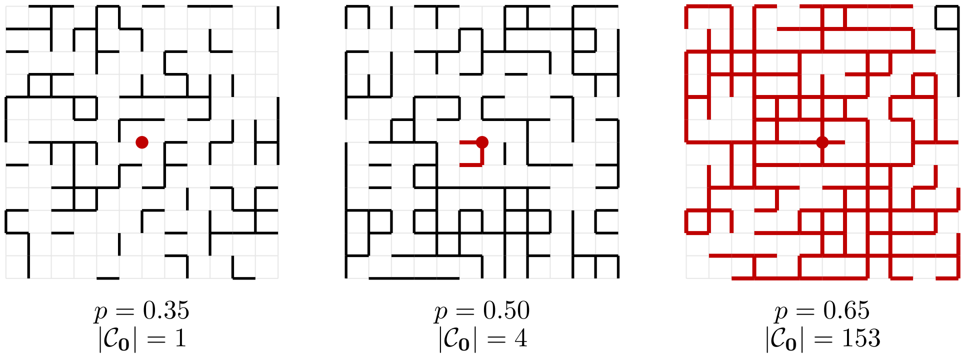


Figure 1: Samples of Bernoulli bond percolation in a 12×12 window of $\mathbb{Z}^2$ at three parameters, drawn from one seed. The open cluster of the marked origin is thickened, and its size within the window is printed underneath. At $p = 0.35$ the cluster is a small island, and at $p = 0.65$ it already spans the window. The middle picture, at the critical probability $p_c(2) = 1/2$, is the one these notes are about, and the eye cannot tell from it whether the cluster would continue forever.

Proof. Each event A_n asserts the existence of an open path, so each A_n is increasing, and therefore $\mathbb{P}_p(A_n) \le \mathbb{P}_q(A_n)$ whenever $p \le q$ by Lemma 1.6. Taking the infimum over n in (1) preserves the inequality, so $\theta(p) \le \theta(q)$.

At $p = 0$ no edge is open, so $\mathcal{C}_{\mathbf{0}} = \{\mathbf{0}\}$ is finite, and therefore $\theta(0) = 0$. □

Since θ is nondecreasing and vanishes at 0, the set of parameters at which it vanishes is an interval containing 0. Its right endpoint is the number the whole subject is organized around.

Definition 1.10 (Critical Probability) The **critical probability** is

$$p_c = p_c(d) := \inf\{p \in [0,1] : \theta(p) > 0\},$$

with the convention that the infimum of the empty set equals 1.

By Lemma 1.9 we have $\theta(p) = 0$ for every $p < p_c$ and $\theta(p) > 0$ for every $p > p_c$. The parameter p_c itself is covered by neither statement, and that single value is what these notes are about.

1.4 The critical probability is strictly between zero and one

For the critical probability to separate two phases it must avoid both endpoints of the parameter interval. Both bounds come from counting paths, and the two counts run in opposite directions: to rule out an infinite cluster we count open paths leaving the origin, and to produce one we count closed circuits that would have to surround it.

Lemma 1.11. (No Infinite Cluster at Small Parameters) *Let* $p < 1/(2d-1)$*. Then* $\theta(p) = 0$.

Proof. The strategy is to bound the percolation probability by the expected number of long open paths from the origin, and then to show that this expectation tends to 0. We divide the proof into three steps: counting the available paths, bounding the expected number of open ones, and comparing with $\theta(p)$.

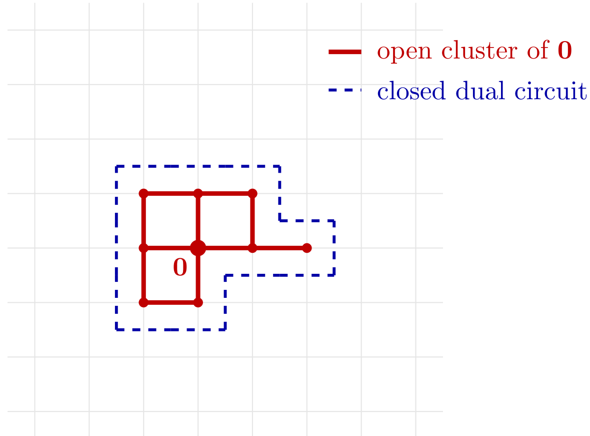


Figure 2: Planar duality, as used in Step 2 of the proof of Lemma 1.12. The open cluster of the origin is thickened, and it is finite exactly when a circuit of closed dual edges, drawn dashed, surrounds the origin. Each dual edge crosses exactly one edge of the lattice and carries its state, so a closed dual circuit blocks every open path out of the region it encloses.

Step 1: count the available paths. Call a path **self-avoiding** when its vertices are distinct. The number of self-avoiding paths of length n starting at the origin is at most $2d(2d-1)^{n-1}$, because the first step has at most $2d$ choices and each later step has at most $2d-1$, the step back along the edge just used being excluded.

Step 2: bound the expected number of open ones. A fixed self-avoiding path of length n uses n distinct edges, so it is open with probability p^n by independence. Summing over the paths counted in Step 1, the expected number of open self-avoiding paths of length n from the origin is at most $2d(2d-1)^{n-1}p^n$.

Step 3: compare with the percolation probability. If $\mathcal{C}_{\mathbf{0}}$ is infinite then for every n it contains a vertex at distance at least n, and a shortest open path from the origin to such a vertex is self-avoiding of length at least n. Hence the event $\{|\mathcal{C}_{\mathbf{0}}| = \infty\}$ is contained in the event that at least one open self-avoiding path of length n from the origin exists, and Markov's inequality applied to the count of Step 2 gives

$$\theta(p) \leq 2d(2d-1)^{n-1}p^n = \frac{2d}{2d-1}\big((2d-1)p\big)^n .$$

When $(2d-1)p < 1$ the right side tends to 0 as n grows, while the left side does not depend on n, so $\theta(p) = 0$. This proves the lemma. $\square$

The other bound uses planar duality, which we describe before using it. Place a dual vertex at the center of each unit square of $\mathbb{Z}^2$, and join two dual vertices by a **dual edge** exactly when their squares share a side. Each edge of $\mathbb{Z}^2$ is crossed by exactly one dual edge, and we declare a dual edge open or closed according to the state of the edge it crosses. A **dual circuit** is a self-avoiding closed path in the dual lattice.

Lemma 1.12. (An Infinite Cluster Exists at Large Parameters) *There is a parameter $p < 1$ with $\theta(p) > 0$.*

Proof. We reduce to the plane and then count dual circuits. The proof has four steps: the reduction, the conversion of finiteness into a dual circuit, the count, and the choice of parameter.

Step 1: reduce to two dimensions. The subgraph of $\mathbb{Z}^d$ induced by the vertex set $\mathbb{Z}^2 \times \{\mathbf{0}\}$ is a copy of $\mathbb{Z}^2$. An infinite open cluster of this subgraph is contained in an infinite open cluster of $\mathbb{Z}^d$, and the restriction of $\mathbb{P}_p$ to the edges of the subgraph is Bernoulli bond percolation on $\mathbb{Z}^2$ with the same parameter. So it is enough to find $p < 1$ with $\theta(p) > 0$ when $d = 2$.

Step 2: convert finiteness into a dual circuit. Work in $\mathbb{Z}^2$ and suppose $\mathcal{C}_\mathbf{0}$ is finite. Every edge joining $\mathcal{C}_\mathbf{0}$ to its complement is closed, so the dual edges crossing those edges are all closed. Let W be the union of $\mathcal{C}_\mathbf{0}$ with the bounded components of its complement, a finite set with connected complement, and consider the edges joining W to its complement. The dual edges crossing them form a self-avoiding closed dual circuit surrounding the origin, which is the standard planar duality statement illustrated in Figure 2. Passing to W matters: the dual edges crossing the full edge boundary of $\mathcal{C}_\mathbf{0}$ form in general several circuits, one around each hole, and it is the outermost one that is wanted. Hence the event that $\mathcal{C}_\mathbf{0}$ is finite is contained in the event that some closed dual circuit surrounds the origin.

Step 3: count the dual circuits. A dual circuit of length n that surrounds the origin must cross the positive horizontal axis, and every vertex of the circuit lies within distance n of the origin, so the crossing occurs at one of at most n dual edges on that axis. Starting from such an edge, the circuit is a self-avoiding dual path with at most three choices at each later step, so the number of dual circuits of length n surrounding the origin is at most $n\,3^{n-1}$. Every such circuit has length at least 4.

Step 4: choose the parameter. A fixed dual circuit of length n is closed with probability $(1-p)^n$, so by Step 3 the expected number of closed dual circuits surrounding the origin is at most $\sum_{n\geq 4} n\,3^{n-1}(1-p)^n$. This series converges when $3(1-p) < 1$, and its value tends to 0 as p tends to 1, so we may fix $p < 1$ for which the value is less than 1. For that p, Markov's inequality shows the probability that some closed dual circuit surrounds the origin is less than 1, so by Step 2 the probability that $\mathcal{C}_\mathbf{0}$ is infinite is positive. This proves the lemma. □

Theorem 1.13. (The Critical Probability Is an Interior Point) *We have* $0 < p_\mathrm{c}(d) < 1$.

Proof. Lemma 1.11 gives $\theta(p) = 0$ for every $p < 1/(2d-1)$, so no parameter below $1/(2d-1)$ belongs to the set appearing in Definition 1.10, and therefore $p_\mathrm{c} \geq 1/(2d-1) > 0$. Lemma 1.12 produces a parameter $p < 1$ with $\theta(p) > 0$, and that parameter does belong to the set, so $p_\mathrm{c} \leq p < 1$. □

In other words, both phases are genuinely present, and the parameter interval really does split into a region where every open cluster is finite and a region where an infinite one exists. What Theorem 1.13 does not say is which of the two regions the critical probability itself belongs to.

1.5 Continuity away from the critical probability

By (1) the percolation probability is an infimum of polynomials, and such an infimum is automatically well behaved from one side. Recording that carefully is what reduces the whole continuity question to the single number $\theta(p_\mathrm{c})$.

Lemma 1.14. (The Percolation Probability Is Right-Continuous) *The function θ is right-continuous at every $p \in [0, 1)$.*

Proof. Fix n. The event A_n is determined by the finitely many edges meeting the ball of radius n about the origin, so $\mathbb{P}_p(A_n)$ is a finite sum of products of the numbers p and $1 - p$, hence a polynomial in p and in particular a continuous function of p.

By (1) the function θ is an infimum of continuous functions, so it is upper semicontinuous. A nondecreasing upper semicontinuous function is right-continuous, and θ is nondecreasing by Lemma 1.9. $\square$

Left-continuity is the harder half. Above the critical probability it follows from the almost sure uniqueness of the infinite open cluster, proved by Aizenman, Kesten, and Newman [2] and given a much shorter proof by Burton and Keane [8]. That uniqueness implies continuity of the percolation probability except possibly at the critical probability is due to van den Berg and Keane [32], and Aizenman, Kesten, and Newman [2] combine the two. Uniqueness is what prevents the percolation probability from being spread across infinitely many competing clusters as the parameter decreases. At p_c the argument has nothing to compare with, because below p_c there is no infinite cluster at all, and left-continuity at p_c is exactly the statement $\theta(p_c) = 0$.

Theorem 1.15. (Continuity Away from the Critical Probability) *The function θ is continuous at every $p \in [0, 1]$ with $p \neq p_c$. Moreover θ is continuous on all of $[0, 1]$ if and only if $\theta(p_c) = 0$.*

Proof. We prove continuity below p_c, then continuity above p_c, and finally the stated equivalence.

Step 1: continuity below the critical probability. If $p < p_c$ then θ vanishes on the whole interval $[0, p_c)$ by Lemma 1.9, and a function that is constant on a neighborhood of p is continuous at p.

Step 2: continuity above the critical probability. If $p_c < p < 1$ then right-continuity at p is Lemma 1.14, and left-continuity at p holds because the infinite open cluster is almost surely unique at every parameter in the supercritical phase. At $p = 1$ only left-continuity is required, and it holds for the same reason. Steps 1 and 2 together give continuity at every parameter other than p_c.

Step 3: the equivalence. Suppose first that $\theta(p_c) = 0$. Then θ vanishes on $[0, p_c]$, and it is right-continuous at p_c by Lemma 1.14, so it is continuous at p_c; with Steps 1 and 2 this gives continuity on $[0, 1]$. Conversely, suppose θ is continuous on $[0, 1]$. Since θ vanishes on $[0, p_c)$, continuity at p_c forces $\theta(p_c) = \lim_{p \uparrow p_c} \theta(p) = 0$. $\square$

Equivalently, the percolation probability has at most one discontinuity, it can sit only at the critical probability, and it would be a jump from the value 0 up to the value $\theta(p_c)$. Deciding whether that jump occurs is the same as computing one number.

1.6 What is known dimension by dimension

The number $\theta(p_c)$ is known to vanish at both ends of the dimension range, by two methods that share nothing with each other.

In two dimensions the critical probability equals $1/2$ and the percolation probability vanishes there. Harris [23] proved in 1960 that $\theta(1/2) = 0$, using the correlation inequality that

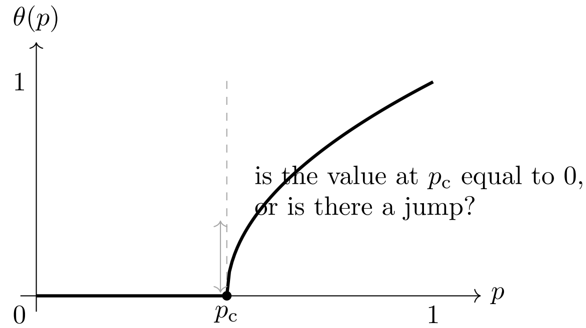


Figure 3: The percolation probability vanishes below the critical probability and is positive above it. By Theorem 1.15 the only possible discontinuity sits at p_c, so the entire question is whether the curve returns to the horizontal axis as the parameter decreases to p_c. The shape drawn above p_c is schematic; only the vanishing below p_c and the positivity above it are asserted here.

now carries his name together with planar duality, and Kesten [24] completed the picture in 1980 by proving $p_c(2) = 1/2$, resting on the crossing estimates proved independently by Russo [29] and Seymour and Welsh [31]; the book-length account is Kesten [25]. For site percolation in the plane the corresponding statement, that the percolation probability is continuous on the whole parameter interval, is due to Russo [30]. Both steps use duality in an essential way, and duality is available only in the plane, so neither argument survives into three dimensions.

In high dimensions the lace expansion applies. Hara and Slade [21] proved $\theta(p_c) = 0$ in high dimensions, and above six dimensions for a class of models with spread-out steps; together with Hara and Slade [22] the threshold for the nearest-neighbor model is $d \geq 19$, although the computations behind that number were never published, as Fitzner and van der Hofstad [14] record. They bring the threshold down to $d \geq 11$. The method compares the probability that two vertices are connected with the corresponding quantity for simple random walk, and the comparison needs enough room for two independent walks to avoid each other, which is what forces the dimension to be large.

The dimensions $3 \leq d \leq 10$ were left open by both methods and remained so for decades, as recorded in Grimmett [19, §8.3] and in Fitzner and van der Hofstad [14, §1.1]. For site percolation the gap was wider still. Writing in August 2026, Cerf [9] calls $\theta(p_c) = 0$ for site percolation in dimensions three and above the main open question of the subject, proves that at p_c either the percolation probability vanishes or the tail of the finite cluster size fails to be a stretched exponential, and names the inequality of Kozma and Nitzan [27] as the route that would settle it. Sharpness of the phase transition, which these notes do not use, was proved independently by Aizenman and Barsky [3] and Menshikov [28], with short modern proofs by Duminil-Copin and Tassion [11] and Duminil-Copin and Tassion [12]; Duminil-Copin [10] surveys what is known at and near the critical point. The standard references for this section are Grimmett [19], Bollobás and Riordan [6] and Friedli and Velenik [16]. Section 2 develops the correlation inequalities that everything here rests on, beginning with the one Harris used. Section 3 then explains a route to the middle dimensions that passes through a statement about finite weighted graphs with no lattice in it at all, and Sections 4 and 5 prove that statement.

2 Correlation Inequalities for Percolation

Percolation has almost no exact computations in it, so nearly every argument is an inequality, and a surprising number of them come from one idea: events that are both helped by opening edges tend to occur together. Harris [23] proved the first such statement in 1960 and used it to show that the percolation probability vanishes at 1/2 in the plane. Stronger and more flexible versions have been developed since, and this section builds the ones these notes need, starting from nothing.

There are two results to reach. The first is Harris's inequality: two increasing events are positively correlated, so the probability that both occur is at least the product of their probabilities. Section 2.1 proves it by induction on the number of edges. Section 2.2 proves it again, this time from the four functions theorem of Ahlswede and Daykin [1], which compares four sums at once and which we need in its own right later.

Harris's inequality is not enough for what follows. The argument these notes are aimed at compares the cluster of one vertex with a set of other vertices while forbidding that cluster from reaching somewhere else, and the forbidding is not optional: it is what keeps the estimate from degrading as that set grows. Conditioning on such an event destroys the independence of the edges, so Harris's inequality does not apply to the conditioned measure. Section 2.3 shows that its conclusion fails as well, by exhibiting two increasing functions that become negatively correlated.

The second result is what survives. Positive correlation does survive this conditioning for increasing functions of the cluster of the conditioned vertex. We prove it in the form where the function may read the whole set of open elements inside the cluster and not only the vertices the cluster occupies. That form contains the other, since the vertices are recovered from the open elements, and it is the form the later sections use. Section 2.4 introduces that set, together with the two facts about exploring a cluster that the proof consumes, and Section 2.5 proves the inequality. For graphs this is the theorem of van den Berg, Häggström, and Kahn [33]; the form proved here, for an arbitrary finite family of independent elements, is ours. It is the engine of everything in Sections 4 and 5.

2.1 Increasing functions and the Harris inequality

Throughout this section $n \geq 1$ is an integer and we work on the finite cube $\{0,1\}^n$ rather than on the lattice; the lattice statements follow by a limit that we record at the end of the section. Order the cube pointwise: write $x \leq y$ when $x_i \leq y_i$ for every i with $1 \leq i \leq n$.

Definition 2.1 (Increasing Function) Let $n \geq 1$ be an integer. A function $f : \{0,1\}^n \to \mathbb{R}$ is **increasing** when $f(x) \leq f(y)$ for all $x \leq y$, and **decreasing** when $-f$ is increasing.

An event is increasing exactly when its indicator function is an increasing function, so Definition 2.1 extends Definition 1.4 from events to functions.

Let $p_1, \ldots, p_n$ be numbers in $[0,1]$ and let μ denote the product measure on $\{0,1\}^n$ under which the coordinates are independent and the i-th coordinate equals 1 with probability p_i. Allowing the parameters to differ across coordinates costs nothing in the proof and is what we need later, so we carry the general case from the start.

The one-dimensional case is a classical rearrangement inequality, and the whole theorem is built from it.

Lemma 2.2. (Positive Correlation in One Coordinate) *Let $p \in [0,1]$ and let f and g be increasing functions on $\{0,1\}$. Let X be a random variable with $\mathbb{P}(X=1)=p$ and $\mathbb{P}(X=0)=1-p$. Then*

$$\mathbb{E}\big[f(X)g(X)\big] \geq \mathbb{E}\big[f(X)\big]\mathbb{E}\big[g(X)\big]\,.$$

Proof. Let X and Y be independent with the same law. Since f and g are both increasing on the two-point set, the product $\big(f(X)-f(Y)\big)\big(g(X)-g(Y)\big)$ is a product of two numbers with the same sign, so it is nonnegative, and therefore its expectation is nonnegative. Expanding,

$$0 \leq \mathbb{E}\big[(f(X)-f(Y))(g(X)-g(Y))\big] = 2\mathbb{E}\big[f(X)g(X)\big] - 2\mathbb{E}\big[f(X)\big]\mathbb{E}\big[g(X)\big]\,,$$

where we used that X and Y are independent with the same law to identify the cross terms. Dividing by 2 gives the inequality. □

Theorem 2.3. (Harris Inequality) *Let $n \geq 0$ be an integer, let $p_1,\dots,p_n$ be numbers in $[0,1]$, and let μ be the product measure on $\{0,1\}^n$ with these parameters. If f and g are increasing functions on $\{0,1\}^n$, then*

$$\int fg\,\mathrm{d}\mu \geq \int f\,\mathrm{d}\mu \int g\,\mathrm{d}\mu\,.$$

In particular, if A and B are increasing events then $\mu(A\cap B) \geq \mu(A)\mu(B)$.

The inequality is due to Harris [23]. It was later derived in much greater generality by Fortuin, Kasteleyn, and Ginibre [15], for measures on a distributive lattice satisfying a positivity condition, and on a product space the two statements agree; the lattice form is due to Kleitman [26].

Proof. We argue by induction on the number of coordinates n, that is, we prove for every $n \geq 1$ that the displayed inequality holds for all increasing f and g on $\{0,1\}^n$ and all parameter vectors.

Base case. For $n=0$ the cube is a single point and both sides are the same number. For $n=1$ the assertion is Lemma 2.2.

Inductive step, first part: condition on the last coordinate. Suppose $n \geq 2$ and the assertion holds for $n-1$ coordinates. Write a point of $\{0,1\}^n$ as (x,t) with $x \in \{0,1\}^{n-1}$ and $t \in \{0,1\}$, and for $t \in \{0,1\}$ define

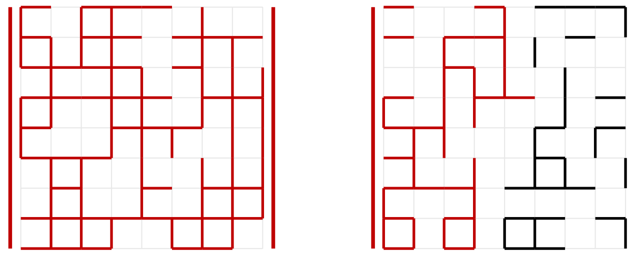
$$F(t) := \int f(x,t)\,\mathrm{d}\mu_{n-1}(x), \qquad G(t) := \int g(x,t)\,\mathrm{d}\mu_{n-1}(x)\,,$$

where μ_{n-1} is the product measure on the first $n-1$ coordinates. For each fixed t the functions $x \mapsto f(x,t)$ and $x \mapsto g(x,t)$ are increasing on $\{0,1\}^{n-1}$, so the induction hypothesis gives

$$\int f(x,t)g(x,t)\,\mathrm{d}\mu_{n-1}(x) \geq F(t)G(t) \qquad \text{for } t \in \{0,1\}\,. \tag{2}$$

Inductive step, second part: the outer coordinate. The functions F and G are increasing on $\{0,1\}$: if $t \leq t'$ then $f(x,t) \leq f(x,t')$ for every x because f is increasing, and integrating preserves the inequality. Let T denote a random variable with $\mathbb{P}(T=1)=p_n$. Integrating (2) against the law of T and then applying Lemma 2.2 to F and G,

$$\int fg\,\mathrm{d}\mu = \mathbb{E}\Big[\int f(x,T)g(x,T)\,\mathrm{d}\mu_{n-1}(x)\Big] \geq \mathbb{E}\big[F(T)G(T)\big] \geq \mathbb{E}\big[F(T)\big]\mathbb{E}\big[G(T)\big]\,,$$



both crossings occur	only the horizontal crossing

Figure 4: Harris's inequality in the case of two crossing events. Both the event that an open path crosses the box from left to right and the event that one crosses from bottom to top become easier when edges are opened, so the two are positively correlated and the probability of seeing both is at least the product of the two probabilities. The left panel shows a configuration where both occur, and the right panel one where only the horizontal crossing occurs.

and the right side equals $\int f \,\mathrm{d}\mu \int g \,\mathrm{d}\mu$ by Fubini's theorem. This completes the induction.

The statement for events follows by taking f and g to be the indicator functions of A and B, which are increasing because A and B are. □

The lattice version follows from the finite one by conditioning. Let A and B be increasing events in $\mathcal{F}_d$ and let $\mathcal{F}_n$ be the σ-algebra generated by the edges meeting the ball of radius n about the origin. The conditional expectations $\mathbb{P}_p(A \mid \mathcal{F}_n)$ and $\mathbb{P}_p(B \mid \mathcal{F}_n)$ are increasing functions of those finitely many coordinates, since averaging an increasing function over the remaining coordinates preserves monotonicity, so Theorem 2.3 applies to them on the finite cube and gives

$$\mathbb{E}_p\big[\mathbb{P}_p(A \mid \mathcal{F}_n)\,\mathbb{P}_p(B \mid \mathcal{F}_n)\big] \ge \mathbb{P}_p(A)\,\mathbb{P}_p(B)\,.$$

By the martingale convergence theorem the two conditional expectations converge almost surely to $\mathbf{1}_A$ and $\mathbf{1}_B$ as n grows, and they are bounded by 1, so the left side converges to $\mathbb{P}_p(A \cap B)$ by dominated convergence. We use the lattice form without further comment.

2.2 The four functions theorem

Harris's inequality compares two integrals. The theorem in this section, due to Ahlswede and Daykin [1], compares four, and it is the flexible tool from which the later arguments are built. For $x, y \in \{0,1\}^n$ write $x \vee y$ for the coordinatewise maximum and $x \wedge y$ for the coordinatewise minimum.

Theorem 2.4. (Four Functions Theorem) *Let $n \ge 0$ be an integer and let f_1, f_2, f_3, f_4 be nonnegative functions on $\{0,1\}^n$ satisfying*

$$f_1(x)f_2(y) \le f_3(x \vee y)f_4(x \wedge y) \qquad \text{for all } x, y \in \{0,1\}^n\,. \tag{3}$$

Then

$$\Big(\sum_x f_1(x)\Big)\Big(\sum_y f_2(y)\Big) \le \Big(\sum_x f_3(x)\Big)\Big(\sum_y f_4(y)\Big)\,,$$

where each sum is over all of $\{0,1\}^n$.

Proof. We induct on n. The strategy is that the hypothesis (3) is stable under summing out one coordinate, so the inductive step is a verification that the four one-coordinate sums again satisfy the hypothesis.

Base case. For $n = 0$ the cube is a single point and the hypothesis at that point is the conclusion. Let $n = 1$ and write $a_i = f_i(0)$ and $b_i = f_i(1)$. Applying (3) to the four pairs (x, y) gives

$$a_1 a_2 \le a_3 a_4, \qquad b_1 b_2 \le b_3 b_4, \qquad a_1 b_2 \le b_3 a_4, \qquad b_1 a_2 \le b_3 a_4\,,$$

and we must show $(a_1 + b_1)(a_2 + b_2) \le (a_3 + b_3)(a_4 + b_4)$. Abbreviate $C = b_3 a_4$ and $D = a_3 b_4$, and put $u = a_1 b_2$ and $v = b_1 a_2$, so that the last two displayed inequalities read $u \le C$ and $v \le C$. Expanding the left side,

$$(a_1 + b_1)(a_2 + b_2) = a_1 a_2 + b_1 b_2 + u + v\,, \tag{4}$$

while $(a_3 + b_3)(a_4 + b_4) = a_3 a_4 + b_3 b_4 + C + D$. Since the first two terms of (4) are already bounded by $a_3 a_4$ and $b_3 b_4$, it suffices to prove $u + v \le C + D$.

If $C = 0$ then $u = v = 0$ and there is nothing to prove, so assume $C > 0$. If $u = 0$ then $u + v = v \le C \le C + D$, so assume $u > 0$ as well. Multiplying the first two displayed inequalities gives

$$uv = (a_1 a_2)(b_1 b_2) \le (a_3 a_4)(b_3 b_4) = CD\,.$$

Suppose first that $u \ge D$. Then $u \le C$ and, from $uv \le CD$, also $v \le CD/u$, so

$$u + v \le u + \frac{CD}{u} \le C + D\,,$$

the last step because $(u - C)(u - D) \le 0$ for $D \le u \le C$, which rearranges to $u^2 + CD \le (C + D)u$. Suppose instead that $u < D$. Then $v \le C$ gives $u + v < D + C$. In both cases $u + v \le C + D$, which proves the base case.

Inductive step. Suppose $n \ge 2$ and the theorem holds for $n - 1$ coordinates. Write a point of $\{0, 1\}^n$ as (x, t) with $x \in \{0, 1\}^{n-1}$ and $t \in \{0, 1\}$, and define functions on $\{0, 1\}^{n-1}$ by

$$\widetilde{f}_i(x) := f_i(x, 0) + f_i(x, 1) \qquad \text{for } 1 \le i \le 4\,.$$

We claim the four functions $\widetilde{f}_1, \widetilde{f}_2, \widetilde{f}_3, \widetilde{f}_4$ satisfy the hypothesis (3) on $\{0, 1\}^{n-1}$. Fix x and y in $\{0, 1\}^{n-1}$ and consider the four functions on the single coordinate t given by

$$g_1(t) = f_1(x, t), \quad g_2(t) = f_2(y, t), \quad g_3(t) = f_3(x \vee y, t), \quad g_4(t) = f_4(x \wedge y, t)\,.$$

For $s, t \in \{0, 1\}$ the hypothesis on $\{0, 1\}^n$ applied to the points (x, s) and (y, t) reads

$$f_1(x, s) f_2(y, t) \le f_3(x \vee y, s \vee t) f_4(x \wedge y, s \wedge t)\,,$$

which says exactly that g_1, g_2, g_3, g_4 satisfy (3) in one coordinate. The base case therefore gives $\widetilde{f}_1(x)\,\widetilde{f}_2(y) \le \widetilde{f}_3(x \vee y)\,\widetilde{f}_4(x \wedge y)$, which is the claim.

Applying the induction hypothesis to $\widetilde{f}_1, \ldots, \widetilde{f}_4$ and noting that $\sum_x \widetilde{f}_i(x) = \sum_{(x,t)} f_i(x, t)$ completes the induction. ☐

Harris's inequality is a special case of Theorem 2.4, and the choice of four functions that produces it is the same one used later under conditioning, in Lemma 2.13.

Corollary 2.5. (Harris Inequality from Four Functions) *Let $n \geq 1$ be an integer, let μ be a product measure on $\{0,1\}^n$, and let f and g be increasing functions. Then $\int fg \, \mathrm{d}\mu \geq \int f \, \mathrm{d}\mu \int g \, \mathrm{d}\mu$.*

Proof. Write $\mu(x)$ for the mass of the point x and set

$$f_1 = f\mu, \qquad f_2 = g\mu, \qquad f_3 = fg\mu, \qquad f_4 = \mu\,.$$

We check the hypothesis (3). A product measure satisfies $\mu(x)\mu(y) = \mu(x \vee y)\mu(x \wedge y)$, since for each coordinate the two sides record the same pair of values. Since f and g are increasing and nonnegative, $f(x) \leq f(x \vee y)$ and $g(y) \leq g(x \vee y)$. Multiplying,

$$f_1(x)f_2(y) = f(x)g(y)\mu(x)\mu(y) \leq f(x \vee y)g(x \vee y)\mu(x \vee y)\mu(x \wedge y) = f_3(x \vee y)f_4(x \wedge y)\,.$$

Theorem 2.4 now gives $\left(\int f \, \mathrm{d}\mu\right)\left(\int g \, \mathrm{d}\mu\right) \leq \left(\int fg \, \mathrm{d}\mu\right) \cdot 1$, which is the claim for nonnegative f and g.

For increasing f and g that are not nonnegative, note that the cube is finite, so f and g are bounded; choose a constant M with $f + M \geq 0$ and $g + M \geq 0$. Both $f + M$ and $g + M$ are increasing and nonnegative, so the case already proved applies to them, and adding a constant to either argument leaves a covariance unchanged. □

2.3 Why the Harris inequality is not enough

We first locate where the unconditional inequality fails, since the shape of the failure dictates the shape of the repair.

Fix a vertex x, a set Y of vertices, and consider the event $\{x \nleftrightarrow Y\}$ that x is connected to no vertex of Y. This event is decreasing: opening more edges can only create connections. Conditioning on it therefore biases the configuration downward, and the conditioned measure is no longer a product measure.

Example 2.6 Take the graph with three vertices x, y and z and the two edges xz and zy, each open with probability $1/2$, and let $Y = \{y\}$. Under the unconditioned measure the two edges are independent. Conditioned on $\{x \nleftrightarrow Y\}$, which here is the event that the two edges are not both open, the edges are no longer independent: the conditional probability that xz is open is $1/3$, and given that xz is open the conditional probability that zy is open is 0. So conditioning on an avoidance event destroys independence, and Theorem 2.3 does not apply to the conditioned measure.

The example does more than show that the proof of Theorem 2.3 cannot be reused. Take f to be the indicator that the edge xz is open and g the indicator that zy is open, both increasing functions of the configuration. Conditioned on $\{x \nleftrightarrow Y\}$ their expectations are each $1/3$ while the expectation of their product is 0, so they are negatively correlated under the conditioned measure. Positive correlation therefore fails outright for arbitrary increasing functions, and what survives the conditioning is the statement restricted to functions of the cluster of x, which is the subject of the next section.

2.4 Hypergraphs, and exploring a cluster

Everything from here on is proved for a model more general than percolation on a graph, in which an open element may join more than two vertices at once. Section 1.1 gave the reason.

The generality costs nothing in the proofs below, and it is what lets the same argument cover site percolation.

Definition 2.7 (Independent Hyperedges) Let U be a finite vertex set and E a finite set of **hyperedges**. Each $e \in E$ carries an **incidence set** $\iota(e) \subseteq U$ with at least two elements, and is **open** with probability p_e, independently of the others. Two vertices are **connected** when a chain of open hyperedges joins them, consecutive hyperedges sharing a vertex. For $S \subseteq U$ write C_S for the union of the connected components meeting S.

Bond percolation is the case in which every incidence set has two elements. Section 6 shows that site percolation is another case, and that it is not a case of bond percolation.

Conditioning on what a cluster does requires remembering slightly more than the set of vertices it occupies.

Definition 2.8 (The Open Hyperedges of a Cluster) For $S \subseteq U$ write

$$\mathcal{K}_S = \{\, e \in E : e \text{ is open and } \iota(e) \cap C_S \neq \varnothing \,\}$$

for the set of open hyperedges that meet the cluster of S.

The cluster is recovered from $\mathcal{K}_S$ together with S, by

$$C_S = S \cup \bigcup_{e \in \mathcal{K}_S} \iota(e)\,,$$

since a vertex of C_S outside S lies on an open hyperedge meeting C_S, and conversely the incidence set of such a hyperedge lies in C_S. If $x \in S$ then the cluster of x is recovered as well, by taking the component of x in the hyperedges of $\mathcal{K}_S$. The set C_S by itself does not determine that cluster, and this already fails when every incidence set has two elements.

Proposition 2.9. (The Cluster of a Set Does Not Determine the Cluster of a Member) *There are a graph, a set S of vertices, a vertex $x \in S$, and two configurations giving S the same C_S but giving x different clusters.*

Proof. Take the path on x, y, z and let $S = \{x, z\}$. If both edges are open then $C_S = \{x, y, z\}$ and the cluster of x is $\{x, y, z\}$. If xy is open and yz is closed then $C_S = \{x, y, z\}$ again, since z belongs to S and y is reached from x, while the cluster of x is now $\{x, y\}$. □

This is why the inequalities below are stated for $\mathcal{K}_S$ rather than for C_S. These sets are ordered by inclusion, and C_S grows with them.

We use one bookkeeping device throughout. For $Z \subseteq U$ write E_Z for the set of hyperedges meeting Z. If $I \subseteq E_Z$ is the set of open ones, its **trace** outside Z is

$$T_Z(I) = \bigcup_{e \in I} \bigl(\iota(e) \setminus Z\bigr)\,. \tag{5}$$

Lemma 2.10. (The Trace Is Monotone, and Almost a Lattice Map) *For all $I, J \subseteq E_Z$,*

$$T_Z(I \cup J) = T_Z(I) \cup T_Z(J), \qquad T_Z(I \cap J) \subseteq T_Z(I) \cap T_Z(J)\,.$$

Proof. The first identity is immediate from (5), since a union of unions is the union. For the second, if $e \in I \cap J$ then $\iota(e) \setminus Z$ is contained in both $T_Z(I)$ and $T_Z(J)$. The inclusion may be strict, because two different hyperedges, one in I only and one in J only, can contribute the same vertex. □

Lemma 2.11. (Exploration Identity) *Let $Z \subseteq N \subseteq U$. Condition on the states of the hyperedges in E_Z and let $I \subseteq E_Z$ be the open ones. Whenever a cluster contained in $U \setminus Z$ avoids N, deleting Z replaces the avoided set N by $(N \setminus Z) \cup T_Z(I)$, and the hyperedges disjoint from Z retain their original independent laws.*

Proof. A path that enters Z must use an open hyperedge meeting Z, that is, a member of I, and the last vertex it occupies before leaving Z lies in the trace (5). Conversely every vertex of the trace is joined to Z by the open hyperedge that put it there. So avoiding N from outside Z is the same as avoiding $(N \setminus Z) \cup T_Z(I)$ in the model with Z deleted. The remaining coordinates are disjoint from E_Z, so exposing E_Z leaves their product law unchanged. $\square$

Conditioning on a whole cluster leaves everything away from it untouched. This is the second bookkeeping fact used throughout, and it is the reason conditioning on one cluster can be iterated.

Lemma 2.12. (Conditioning on a Cluster Leaves the Rest Unbiased) *Let $S \subseteq U$ and let $A \subseteq E$ be a set of hyperedges such that $\mathbb{P}(\mathcal{K}_S = A) > 0$. Write W for the union of S with the incidence sets of the members of A. Conditionally on $\{\mathcal{K}_S = A\}$ the hyperedges whose incidence sets are disjoint from W are independent, each open with its original probability p_e.*

Proof. On $\{\mathcal{K}_S = A\}$ we have $C_S = W$: every member of A is open and meets C_S, so its incidence set lies in C_S, and conversely a vertex of C_S outside S is reached by a chain of open hyperedges, each of which meets C_S and is therefore recorded in A. Consequently a hyperedge is open and meets W exactly when it belongs to A, so

$$\{\mathcal{K}_S = A\} = \bigcap_{e \in A} \{e \text{ open}\} \cap \bigcap_{e \notin A,\ \iota(e) \cap W \neq \varnothing} \{e \text{ closed}\}.$$

For the reverse inclusion, use $\mathbb{P}(\mathcal{K}_S = A) > 0$: some configuration has $\mathcal{K}_S = A$, and in it every vertex met by a hyperedge of A is joined to S through hyperedges of A alone, since every open hyperedge meeting C_S belongs to A. So if all of A is open then $W \subseteq C_S$, and if in addition every hyperedge outside A meeting W is closed then no path leaves W, so $C_S = W$ and the open hyperedges meeting C_S are exactly A.

The displayed event is therefore determined by the states of the hyperedges meeting W. The remaining hyperedges are independent of those, so conditioning does not change their joint law. $\square$

2.5 Conditional association

We can now prove that conditioning on an avoidance event preserves positive correlation. Write

$$K_U(f; X) = \mathbb{E}\big[f(\mathcal{K}_s)\,;\, C_s \cap X = \varnothing\big]$$

for a nonnegative increasing function f of $\mathcal{K}_s$, where s is a vertex.

Lemma 2.13. (One-Cluster Inequality) *Let $s \in U$, let $X, Y \subseteq U$, and let f and g be nonnegative increasing functions of $\mathcal{K}_s$. Then*

$$K_U(f; X)\, K_U(g; Y) \;\leq\; K_U(fg; X \cap Y)\, K_U(1; X \cup Y). \tag{6}$$

Proof. We induct on the number of vertices of U. If $s \in X \cup Y$ then one factor on the right is zero and one on the left is zero, so assume $s \notin X \cup Y$. Put $Z = X \cap Y$.

The case $Z = \varnothing$. Write A_X for the event $C_s \cap X = \varnothing$ and A_Y for $C_s \cap Y = \varnothing$. The functions $f(\mathcal{K}_s)$ and $g(\mathcal{K}_s)$ are increasing in the hyperedge coordinates and the indicators of A_X and A_Y are decreasing, so Theorem 2.3 gives four inequalities:

$$\mathbb{E}[f; A_X] \le \mathbb{E}[f]\,\mathbb{P}(A_X), \qquad \mathbb{E}[g; A_Y] \le \mathbb{E}[g]\,\mathbb{P}(A_Y),$$

$$\mathbb{E}[f]\,\mathbb{E}[g] \le \mathbb{E}[fg], \qquad \mathbb{P}(A_X)\,\mathbb{P}(A_Y) \le \mathbb{P}(A_X \cap A_Y)\,.$$

The first two apply the theorem to an increasing function and a decreasing one, which is the theorem applied to the increasing pair f and the negative of the indicator. The third applies it to two increasing functions and the fourth to two decreasing ones. Multiplying the first two and then using the third and the fourth,

$$\mathbb{E}[f; A_X]\,\mathbb{E}[g; A_Y] \le \mathbb{E}[fg]\,\mathbb{P}(A_X \cap A_Y)\,.$$

Since $X \cap Y = \varnothing$ the left side is $K_U(f; X)K_U(g; Y)$, while $\mathbb{E}[fg] = K_U(fg; \varnothing)$ and $\mathbb{P}(A_X \cap A_Y) = K_U(1; X \cup Y)$. This is (6).

The inductive step. Suppose $Z \neq \varnothing$. Expose the states of the hyperedges in E_Z and delete Z; by Lemma 2.11 each of the four factors in (6) becomes a sum over the exposed open set $I \subseteq E_Z$, with the avoided set replaced by its trace. For two exposed states I and J apply the induction hypothesis in $U \setminus Z$, which has fewer vertices, with avoided sets

$$(X \setminus Z) \cup T_Z(I) \qquad \text{and} \qquad (Y \setminus Z) \cup T_Z(J)\,.$$

By Lemma 2.10 the union of these two sets is $((X \cup Y) \setminus Z) \cup T_Z(I \cup J)$, and their intersection contains $((X \cap Y) \setminus Z) \cup T_Z(I \cap J)$. Enlarging an avoided set can only decrease an avoidance probability, so the induction hypothesis gives, for each pair I and J, the pointwise inequality indexed by I, J, $I \cap J$ and $I \cup J$ that the four functions theorem requires.

Let $\pi(I)$ be the probability that the exposed open set is exactly I. Since the hyperedges are independent, $\pi(I)\pi(J) = \pi(I \cap J)\,\pi(I \cup J)$. Multiplying the pointwise inequality by these weights puts it in the form of the hypothesis (3), so Theorem 2.4, applied on the Boolean lattice of subsets of E_Z, gives (6) after summing over I and J. $\square$

Corollary 2.14. (Conditional Association Given an Avoidance Event) *Let $s \in U$ and $X \subseteq U$ with $\mathbb{P}(C_s \cap X = \varnothing) > 0$. Then increasing functions of $\mathcal{K}_s$ are positively correlated under the conditional law given $C_s \cap X = \varnothing$.*

Proof. Take $X = Y$ in Lemma 2.13 and divide by $\mathbb{P}(C_s \cap X = \varnothing)^2$, which is positive. This gives the assertion for nonnegative f and g. For real-valued increasing f and g, add a constant to each to make them nonnegative, which is possible because the state space is finite; adding a constant to either argument leaves a covariance unchanged. $\square$

The restriction to functions of $\mathcal{K}_s$ cannot be dropped. Example 2.6 exhibits two increasing functions of the configuration that are negatively correlated under such a conditioning.

2.6 Further consequences of cluster exploration

Six consequences of the results just proved are collected here. They are the tools the proof of Theorem 4.9 consumes, and nothing else in these notes uses them, so a reader may go on to Section 3 and return when that proof begins. The first two are known for graphs and are proved here in the general model; the last four are due to Anthropic.

One piece of notation runs through the six. For $W \subseteq U$ write $\mathbb{P}_W$ and $\mathbb{E}_W$ for the model induced on W, in which a hyperedge is retained only when its incidence set is contained in W, and write C_S^W and $\mathcal{K}_S^W$ for the cluster of S there and for the open hyperedges of that cluster. If a named vertex lies outside W its cluster is empty and every connection to it fails, and $C_\varnothing^W = \varnothing$.

The first of the six compares two clusters rather than one. The conditioning forces them apart, so growing one shrinks the other, and the inequality records that the two effects still point the same way. For graphs this is a conditional correlation inequality of van den Berg, Häggström, and Kahn [33], and the resampling argument below is theirs.

Lemma 2.15. (Two Clusters Given Disconnection) *Let S and T be disjoint nonempty subsets of U with $\mathbb{P}(C_S \cap T = \varnothing) > 0$. Conditional on $C_S \cap T = \varnothing$, two nonnegative functions of $(\mathcal{K}_S, \mathcal{K}_T)$ that are increasing in the first argument and decreasing in the second are positively correlated.*

Proof. Let F and G be the two functions. Condition first on $\mathcal{K}_S = J$, and let $K = C_S$ be the support of J. Lemma 2.12 says that the hyperedges contained in $U \setminus K$ retain their product law. With J fixed, both F and G are decreasing functions of $\mathcal{K}_T$, so the Harris inequality, applied to their negatives, gives

$$\mathbb{E}[FG \mid \mathcal{K}_S = J] \geq \mathbb{E}[F \mid \mathcal{K}_S = J]\,\mathbb{E}[G \mid \mathcal{K}_S = J].$$

Both conditional means on the right are increasing functions of J on the set of values $\mathcal{K}_S$ can take. Indeed, suppose $J \subseteq J'$. The support of J' contains the support of J. Couple the two induced models outside these supports by using the same hyperedge states. Removing the larger support can only decrease $\mathcal{K}_T$. At the same time the first argument of F and G has increased. Thus both functions, and hence both conditional means, can only increase.

A conditional mean is so far defined only at the values $\mathcal{K}_S$ can take. Extend it to every set A of hyperedges by

$$\overline{m}(A) = \max\bigl(\{m(I) : I \subseteq A \text{ and } m \text{ is defined at } I\} \cup \{0\}\bigr).$$

This extension is increasing and agrees with m wherever m was defined. Corollary 2.14 also holds for a finite source set: adjoin a new vertex s, add a deterministic open hyperedge with incidence set $S \cup \{s\}$, and apply that corollary to the cluster of s. Consequently the two conditional means are positively correlated under the law of $\mathcal{K}_S$ given $C_S \cap T = \varnothing$. Averaging the displayed conditional Harris inequality proves the lemma. $\square$

The next lemma lets a cluster be explored one hyperedge at a time before the Harris inequality is applied. Without it, exposing a cluster would destroy the product structure that inequality needs. Gladkov [17] proved a form of the Harris inequality for decision trees; this is the version the argument below uses.

Lemma 2.16. (Harris Inequality at a Stopping Time) *Let μ be a product Bernoulli measure on finitely many coordinates. Let $\mathcal{T}$ be an adaptive procedure that reveals previously unrevealed coordinates, one at a time, and eventually stops. Write $\mathcal{F}_{\mathcal{T}}$ for the information it has revealed. If f and g are increasing real functions, then*

$$\operatorname{Cov}_\mu\big(\mathbb{E}_\mu[f \mid \mathcal{F}_{\mathcal{T}}], g\big) \geq 0. \tag{7}$$

Proof. Let ω and ω' be independent samples from μ. Run $\mathcal{T}$ on ω, and form a third configuration ξ by taking each revealed coordinate from ω and each unrevealed coordinate from ω'. We first prove

$$\mathbb{E}[f(\omega)g(\xi)] \geq \mathbb{E}[f]\,\mathbb{E}[g]. \tag{8}$$

We induct on the largest number of further queries that the procedure can make. If it stops without a query, then $\xi = \omega'$ and equality holds. Otherwise let i be its first query, let p_i be the opening probability, and let $\mathcal{T}_0$ and $\mathcal{T}_1$ be the procedures followed after the answers 0 and 1. Write f_b and g_b for the restrictions to the branch with coordinate i equal to b. The induction hypothesis in the two branches gives

$$\mathbb{E}[f(\omega)g(\xi)] \geq (1-p_i)\mathbb{E}[f_0]\mathbb{E}[g_0] + p_i\mathbb{E}[f_1]\mathbb{E}[g_1].$$

Since f and g are increasing, $\mathbb{E}[f_0] \leq \mathbb{E}[f_1]$ and $\mathbb{E}[g_0] \leq \mathbb{E}[g_1]$. Lemma 2.2 applied to these two pairs of numbers shows that the last display is at least $\mathbb{E}[f]\mathbb{E}[g]$. This proves (8).

Conditional on $\mathcal{F}_{\mathcal{T}}$, the unrevealed coordinates of ω and ω' are independent and have their original product laws. Therefore

$$\mathbb{E}[f(\omega)g(\xi)] = \mathbb{E}\left[\mathbb{E}[f \mid \mathcal{F}_{\mathcal{T}}]\mathbb{E}[g \mid \mathcal{F}_{\mathcal{T}}]\right] = \mathbb{E}\left[\mathbb{E}[f \mid \mathcal{F}_{\mathcal{T}}]g\right].$$

Together with (8), this is (7). □

The two lemmas just proved combine into the estimate the induction actually uses: what a covariance loses when a whole cluster is deleted from the model.

Lemma 2.17. (Covariance after Exploring a Cluster) *Let $x, v \in U$, let $S \subseteq U$ satisfy $x \in S$ and $v \notin S$, and let f be a nonnegative increasing function of $\mathcal{K}_x$. For $W \subseteq U$, set*

$$H(W) = \operatorname{Cov}_W\big(f(\mathcal{K}_x^W), \mathbf{1}\{v \leftrightarrow S\}\big),$$

and set $H(W) = 0$ if $x \notin W$ or $v \notin W$. Then $H(W) \geq 0$, and for every $B \subseteq W$,

$$\mathbb{E}_W\big[H(W \setminus C_B^W); x \notin C_B^W\big]\,\mathbb{P}_W(v \not\leftrightarrow S) \leq \mathbb{P}_W(v \not\leftrightarrow S \cup B)\,H(W). \tag{9}$$

Proof. The two variables defining $H(W)$ are increasing functions of the hyperedge states, so $H(W) \geq 0$ by Theorem 2.3. If $\mathbb{P}_W(v \not\leftrightarrow S) = 0$, both sides of (9) are zero. We may therefore suppose that this probability is positive.

Explore all of $\mathcal{K}_B$, and let $\mathcal{F}_B$ be the information revealed. Put

$$G = f(\mathcal{K}_x^W), \qquad J = \mathbf{1}\{v \leftrightarrow S\}, \qquad Z = \mathbf{1}\{v \not\leftrightarrow S,\ v \leftrightarrow B\}, \qquad \widehat{G} = \mathbb{E}_W[G \mid \mathcal{F}_B].$$

The law of total covariance and Lemma 2.12 give

$$H(W) = \mathbb{E}_W\big[H(W \setminus C_B^W); x \notin C_B^W\big] + \operatorname{Cov}_W(\widehat{G}, J). \tag{10}$$

On the event $x \notin C_B^W$, the first conditional covariance is the covariance in the model induced on $W \setminus C_B^W$. On the complementary event G has been determined, so that conditional covariance is zero.

The variable $J + Z$ is the indicator of the increasing event $\{v \leftrightarrow S \cup B\}$. Lemma 2.16, applied to the exploration of C_B^W, gives

$$\mathrm{Cov}_W(\widehat{G}, J + Z) \geq 0.$$

The variable Z is $\mathcal{F}_B$-measurable, so $\mathrm{Cov}_W(\widehat{G}, Z) = \mathrm{Cov}_W(G, Z)$.

Let $D = \{v \nleftrightarrow S\}$, $a = \mathbb{P}_W(D)$, and $r = \mathbb{P}_W(Z)$. Conditional on D, the clusters of S and v are disjoint. Apply Lemma 2.15 to G, viewed as an increasing function of $\mathcal{K}_S$, and to $1 - \mathbf{1}\{v \leftrightarrow B\}$, viewed as a decreasing function of $\mathcal{K}_v$. It follows that

$$\mathrm{Cov}_W(G, Z) \leq \frac{r}{a} \mathrm{Cov}_W(G, \mathbf{1}_D) = -\frac{r}{a} H(W).$$

Using this bound in (10) yields

$$\mathbb{E}_W\big[H(W \setminus C_B^W); x \notin C_B^W\big] \leq \Big(1 - \frac{r}{a}\Big) H(W).$$

Finally, $a - r = \mathbb{P}_W(v \nleftrightarrow S \cup B)$. Multiplication by a proves (9). □

The next lemma replaces a connection probability computed in the model that survives a deletion by the corresponding probability computed globally. It is the step where hyperedges have to be tracked by name rather than by the vertices they reach, since by Lemma 2.10 two of them can leave the same trace.

Lemma 2.18. (Comparison after Deleting a Cluster) *Let x, o, v be distinct vertices, let $A \subseteq U$ contain x and contain neither o nor v, and let $F : 2^U \to [0, \infty)$ satisfy $F(W) = 0$ whenever $v \notin W$. Define*

$$q_A(W) = \begin{cases} \mathbb{P}_W(o \leftrightarrow v,\ v \nleftrightarrow A)/\mathbb{P}_W(v \nleftrightarrow A), & \mathbb{P}_W(v \nleftrightarrow A) > 0, \\ 0, & \mathbb{P}_W(v \nleftrightarrow A) = 0. \end{cases}$$

Suppose that, for every $W \subseteq U$ and every $B \subseteq W$,

$$\mathbb{E}_W\big[F(W \setminus C_B^W); x \notin C_B^W\big]\, \mathbb{P}_W(v \nleftrightarrow A) \leq \mathbb{P}_W(v \nleftrightarrow A \cup B)\, F(W). \tag{11}$$

Then every $Y \subseteq U$ satisfies

$$\begin{aligned} &\mathbb{P}(o \leftrightarrow v,\ v \nleftrightarrow A \cup Y)\, \mathbb{E}\big[F(U \setminus C_Y); x \notin C_Y\big] \\ &\qquad \leq \mathbb{P}(v \nleftrightarrow A \cup Y)\, \mathbb{E}\big[q_A(U \setminus C_Y) F(U \setminus C_Y); x \notin C_Y\big]. \end{aligned} \tag{12}$$

Proof. The reason for keeping labels in the following argument is that two hyperedges meeting an exposed set can have the same trace outside that set. Their labels remain distinct even in that case.

The stronger assertion. Let L be a finite label set and attach to each $\ell \in L$ a set of vertices $\alpha(\ell) \subseteq U$. For $Q \subseteq L$ and $W \subseteq U$, write

$$[Q]_W = W \cap \bigcup_{\ell \in Q} \alpha(\ell), \qquad R_Q^W = W \setminus C^W_{[Q]_W}.$$

Define

$$\begin{aligned}
e_W(Q) &= \mathbb{P}_W(o \leftrightarrow v,\ v \not\leftrightarrow A \cup [Q]_W),\\
m_W(Q) &= \mathbb{P}_W(v \not\leftrightarrow A \cup [Q]_W),\\
y_W(Q) &= \mathbb{E}_W\left[F(R_Q^W); x \notin C^W_{[Q]_W}\right],\\
z_W(Q) &= \mathbb{E}_W\left[q_A(R_Q^W)F(R_Q^W); x \notin C^W_{[Q]_W}\right].
\end{aligned}$$

We prove, simultaneously for every finite label set and every such attachment, that

$$e_W(N)y_W(N') \le m_W(N \cup N')z_W(N \cap N') \qquad \text{for all } N, N' \subseteq L. \tag{13}$$

The proof is by strong induction on $|W|$.

If $v \in A \cup [N]_W$ or $o \in [N]_W$, then $e_W(N) = 0$. If $x \in [N']_W$, then $y_W(N') = 0$. If $v \notin W$, the condition on F also gives $y_W(N') = 0$. These cases establish (13), so exclude them below.

Disjoint label sets. Suppose that $N \cap N' = \varnothing$. Apply Lemma 2.13, with source v, to the constant function 1 and the increasing indicator $\mathbf{1}\{o \in C_v\}$. Use the two avoided sets $A \cup [N]_W$ and $A \cup [N']_W$. Their intersection contains A, even when the sets attached to distinct labels overlap. Monotonicity in the avoided set therefore gives

$$m_W(N')e_W(N) \le e_W(\varnothing)m_W(N \cup N'). \tag{14}$$

The assumption (11), with $B = [N']_W$, says

$$y_W(N')m_W(\varnothing) \le m_W(N')F(W).$$

If $m_W(\varnothing) > 0$, multiply this inequality by $e_W(N)$, use (14), and divide by $m_W(\varnothing)$. Since

$$z_W(\varnothing) = q_A(W)F(W) = \frac{e_W(\varnothing)}{m_W(\varnothing)}F(W),$$

the result is (13). If $m_W(\varnothing) = 0$, then $e_W(N) = 0$, and the result is immediate.

A common label. Suppose that $Z = N \cap N'$ is nonempty. A label whose attached set misses W can be discarded. If this makes Z empty, the first case applies. We may thus assume that

$$K = [Z]_W$$

is nonempty. The zero cases already removed imply $x, o, v \notin K$. Expose every hyperedge meeting K, delete K, and let E_K be the set of exposed hyperedge labels. In $W \setminus K$ use the new label set

$$(L \setminus Z) \,\dot\cup\, E_K.$$

Attach $\alpha(\ell) \setminus K$ to an old label ℓ and $\iota(e) \setminus K$ to a new label $e \in E_K$.

For $I, J \subseteq E_K$, put

$$\widehat{N}_I = (N \setminus Z) \,\dot\cup\, I, \qquad \widehat{N}'_J = (N' \setminus Z) \,\dot\cup\, J.$$

The labels form a disjoint union, so

$$\widehat{N}_I \cup \widehat{N}'_J = ((N \cup N') \setminus Z) \,\dot\cup\, (I \cup J), \tag{15}$$

$$\widehat{N}_I \cap \widehat{N}'_J = I \cap J. \tag{16}$$

Let $\pi(I)$ be the product probability that exactly the hyperedges in I are open among those meeting K. Lemma 2.11 and Lemma 2.12 expand the four terms in (13) as

$$e_W(N) = \sum_{I \subseteq E_K} \pi(I)e(I), \qquad y_W(N') = \sum_{J \subseteq E_K} \pi(J)y(J),$$
$$m_W(N \cup N') = \sum_{I \subseteq E_K} \pi(I)m(I), \qquad z_W(N \cap N') = \sum_{I \subseteq E_K} \pi(I)z(I).$$

Here $e(I)$ and $y(J)$ are the corresponding quantities in $W \setminus K$ with sources $\widehat{N}_I$ and $\widehat{N}'_J$. The functions $m(H)$ and $z(H)$ use the sources $((N \cup N') \setminus Z) \,\dot{\cup}\, H$ and H, respectively.

The induction hypothesis in $W \setminus K$, applied to $\widehat{N}_I$ and $\widehat{N}'_J$, and (15)–(16) give

$$e(I)y(J) \le m(I \cup J)z(I \cap J).$$

Independence of the exposed hyperedges gives

$$\pi(I)\pi(J) = \pi(I \cup J)\pi(I \cap J).$$

Thus the four functions

$$\pi(I)e(I), \quad \pi(I)y(I), \quad \pi(I)m(I), \quad \pi(I)z(I)$$

satisfy the hypothesis of Theorem 2.4. That theorem proves (13). Since K is nonempty, this completes the strong induction.

Return to vertex sources. Give each vertex of Y a separate singleton label and take $N = N' = L$. Then $[L]_U = Y$. Substitution in (13) gives exactly (12). ☐

Two disjoint clusters may be exposed in either order. The identity below says that the two orders give the same answer, and it is what lets one vertex be removed from a source set in the last lemma of the six.

Lemma 2.19. (Exchanging the Order of Two Cluster Explorations) *For $A \subseteq W$ and $K \subseteq W$, set*

$$\mu_A^W(K) = \mathbb{P}_W(C_A^W = K).$$

Let $A, B \subseteq W$, and let $K, L \subseteq W$ be disjoint sets satisfying $A \subseteq K$ and $B \subseteq L$. Then

$$\mu_A^W(K)\mu_B^{W\setminus K}(L) = \mu_B^W(L)\mu_A^{W\setminus L}(K). \tag{17}$$

Proof. On the event $C_A^W = K$, every hyperedge meeting both K and $W \setminus K$ is closed. By Lemma 2.12, or by summing that lemma over all values with support K, the hyperedges contained in $W \setminus K$ retain their product law. The left side of (17) is therefore

$$\mathbb{P}_W(C_A^W = K,\ C_B^W = L).$$

Exploring the B-cluster first shows that the right side is the same joint probability. ☐

That last lemma is the exact identity moving the induction from one level to the next.

Lemma 2.20. (Adding One Vertex to a Source) *Let x and z be distinct vertices. Let Y and S be disjoint subsets of U with $x \in S$ and $z \notin Y \cup S$. Let g be a bounded real function of $\mathcal{K}_x$. Put*

$$d_x = \mathbb{P}(x \nleftrightarrow Y), \qquad d_z = \mathbb{P}(z \nleftrightarrow Y \cup S), \qquad c(u) = \mathbb{P}(u \in C_z \mid z \nleftrightarrow Y \cup S),$$

and suppose $d_x d_z > 0$. Conditional on $x \nleftrightarrow Y$, expose C_Y, put $W = U \setminus C_Y$, and denote the law of W by π. For $T \subseteq U$, define

$$r_T^W(u) = \mathrm{Cov}_W\big(g(\mathcal{K}_x^W), \mathbf{1}\{u \leftrightarrow T\}\big).$$

For a set K, first form C_Y^{-K} in the configuration obtained by deleting K and every hyperedge meeting K. In the expression $\mathcal{K}_x^{-C_Y^{-K}}$, restore the hyperedges meeting K and then delete C_Y^{-K} and every hyperedge meeting it. Define

$$\Phi(K) = \mathbb{E}\left[\mathbf{1}\{C_x^{-K} \cap Y = \varnothing\}\left(g(\mathcal{K}_x^{-C_Y^{-K}}) - g(\mathcal{K}_x^{-K})\right)\right]. \tag{18}$$

Then, for every vertex u,

$$\begin{aligned}\mathbb{E}_\pi &\left[r_S^W(u) - r_{S\cup\{z\}}^W(u) - c(u) r_S^W(z)\right] \\ &= \frac{d_z}{d_x}\,\mathrm{Cov}\left(\Phi(C_z), \mathbf{1}\{u \in C_z\} \,|\, z \nleftrightarrow Y \cup S\right).\end{aligned} \tag{19}$$

Proof. For an induced vertex set R containing x, put

$$a(R) = \mathbb{E}_R[g(\mathcal{K}_x^R)].$$

A fixed induced hypergraph. We first prove an identity with W fixed. Suppose W contains x and z, let γ be a real number, and set

$$D = \{z \nleftrightarrow S\}, \qquad J_u = D \cap \{u \in C_z^W\}.$$

The disjoint identities

$$\mathbf{1}\{u \leftrightarrow S \cup \{z\}\} = \mathbf{1}\{u \leftrightarrow S\} + \mathbf{1}_{J_u}, \qquad \mathbf{1}\{z \leftrightarrow S\} = 1 - \mathbf{1}_D$$

give

$$\begin{aligned}&r_S^W(u) - r_{S\cup\{z\}}^W(u) - \gamma r_S^W(z) \\ &\quad = -\,\mathrm{Cov}_W(g(\mathcal{K}_x^W), \mathbf{1}_{J_u}) + \gamma\,\mathrm{Cov}_W(g(\mathcal{K}_x^W), \mathbf{1}_D).\end{aligned}$$

On D, the set $K = C_z^W$ is disjoint from S. Summing Lemma 2.12 over the sets $\mathcal{K}_z^W$ whose incidence sets together with z make up K shows that, conditional on $C_z^W = K$, the hyperedges contained in $W \setminus K$ still have their product law, so

$$\mathbb{E}_W[g(\mathcal{K}_x^W) \mid C_z^W = K] = a(W \setminus K).$$

Expanding the two covariances therefore yields

$$\begin{aligned}&r_S^W(u) - r_{S\cup\{z\}}^W(u) - \gamma r_S^W(z) \\ &\quad = \sum_{\substack{K \subseteq W \\ K\cap S = \varnothing}} \mu_{\{z\}}^W(K)\big(a(W) - a(W \setminus K)\big)\big(\mathbf{1}\{u \in K\} - \gamma\big).\end{aligned} \tag{20}$$

If $z \notin W$, both sides are interpreted as zero.

Expanding Φ. We next expand (18). For every K disjoint from $Y \cup S$, condition on $C_Y^{U\setminus K} = Q$. Exact cluster conditioning gives

$$\Phi(K) = \sum_{\substack{Q \subseteq U\setminus K \\ x \notin Q}} \mu_Y^{U\setminus K}(Q) \left[a(U \setminus Q) - a(U \setminus (K \cup Q))\right]. \tag{21}$$

For the first term in the brackets, the hyperedges meeting K were absent from the exploration of $C_Y^{U\setminus K}$ and retain their independent states. Together with the hyperedges disjoint from $K \cup Q$, they give the product model on $U \setminus Q$. For the second term, the boundary of Q is closed in $U \setminus K$, leaving the product model on $U \setminus (K \cup Q)$. The condition $x \notin Q$ is precisely the indicator in (18).

Exchanging the order. Take $\gamma = c(u)$ in (20). Multiplying the conditional covariance on the right side of (19) by d_z and using (21) gives

$$\sum_{\substack{K \subseteq U \\ K \cap (Y \cup S) = \varnothing}} \mu_{\{z\}}^U(K)\Phi(K)\big(\mathbf{1}\{u \in K\} - c(u)\big).$$

Insert the sum over Q from (21). Lemma 2.19 changes each product of cluster probabilities according to

$$\mu_{\{z\}}^U(K)\mu_Y^{U\setminus K}(Q) = \mu_Y^U(Q)\mu_{\{z\}}^{U\setminus Q}(K).$$

After interchanging the two finite sums, the preceding display becomes

$$\sum_{\substack{Q \subseteq U \\ x \notin Q}} \mu_Y^U(Q) \sum_{\substack{K \subseteq U\setminus Q \\ K \cap S = \varnothing}} \mu_{\{z\}}^{U\setminus Q}(K) \left[a(U \setminus Q) - a(U \setminus (K \cup Q))\right] \cdot \big(\mathbf{1}\{u \in K\} - c(u)\big).$$

Terms with $z \in Q$ are zero and have only been added. By (20), the inner sum is

$$r_S^{U\setminus Q}(u) - r_{S\cup\{z\}}^{U\setminus Q}(u) - c(u) r_S^{U\setminus Q}(z).$$

Finally,

$$\frac{\mu_Y^U(Q)}{d_x}, \qquad x \notin Q,$$

is the law of $U \setminus C_Y$ conditional on $x \nleftrightarrow Y$. Dividing by d_x proves (19). $\square$

3 Gluing Inequalities and the Critical Probability

Suppose we know two things: that a source vertex reaches some vertex of a given set with high probability, and that each vertex of that set reaches a target with high probability. Can we conclude that the source reaches the target? When the set has one element this is the union bound and there is nothing to prove. The difficulty is that the lattice argument of this section produces sets whose size grows without bound, and the union bound degrades linearly in that size. What is needed is a bound that does not.

Kozma and Nitzan [27] proposed such statements as conjectures in 2024 and proved that one of them settles the vanishing of the percolation probability at the critical probability in every dimension. Their reduction is what makes the middle dimensions accessible. It replaces a lattice question, on which duality and the lace expansion both fail, by a combinatorial statement about finite weighted graphs with no lattice in it.

Section 3.1 sets up the problem of connecting through a set and shows how the union bound fails. Section 3.2 states the three gluing inequalities we need to distinguish. Section 3.3 states the reduction and describes its two cases. Section 3.4 records what is left to prove, which is the subject of the two sections that follow.

Multiplicative gluing below is Conjecture 1 of Kozma and Nitzan [27], on page 3 of that paper, and near-one gluing is their Conjecture 3, on page 15; they record that they could neither prove nor disprove the first. The additive form does not appear in their paper. All three are proved in these notes, and we keep them apart because they are not equally easy to reach: the argument of Section 5 produces the multiplicative one, the other two follow from it in a line, and it is the weakest of the three that the lattice application consumes.

3.1 Connecting through a set

Throughout this section the model is the one of Definition 2.7: a finite vertex set, finitely many hyperedges, each open independently with its own probability. Bond percolation on a finite graph is the case in which every incidence set is a pair, and site percolation is another case, as Section 6.1 shows, so everything proved here holds for both. Write o and b for vertices, A for a nonempty set of vertices, and $o \leftrightarrow A$ for the event that o is connected to at least one vertex of A.

The situation to have in mind is the following. We know

$$\mathbb{P}(o \leftrightarrow A) \geq 1 - \delta \qquad \text{and} \qquad \mathbb{P}(a \leftrightarrow b) \geq 1 - \delta \quad \text{for every } a \in A\,, \tag{22}$$

and we want a lower bound on $\mathbb{P}(o \leftrightarrow b)$. The naive route is a union bound over the vertices of the set.

Lemma 3.1. (The Union Bound over A) *In the setting above,*

$$\mathbb{P}(o \not\leftrightarrow b) \leq \mathbb{P}(o \not\leftrightarrow A) + \sum_{a \in A} \mathbb{P}(a \not\leftrightarrow b)\,.$$

Proof. If o is connected to b then no term on the right is needed, so assume the event $\{o \not\leftrightarrow b\}$ occurs. Either o fails to reach A, which is the first term, or o reaches some vertex $a \in A$; in the latter case that a cannot reach b, since otherwise o would reach b by concatenating the two open paths. Hence

$$\{o \not\leftrightarrow b\} \subseteq \{o \not\leftrightarrow A\} \cup \bigcup_{a \in A} \{a \not\leftrightarrow b\}\,,$$

and the claim follows from countable subadditivity. □

Under the hypotheses (22) the lemma gives $\mathbb{P}(o \not\leftrightarrow b) \leq \delta + |A|\delta$, which is useless as soon as $|A| \geq 1/\delta$. The failure is not an artifact of a crude proof: the lemma discards the information that the vertices of the set are alternatives to one another rather than requirements, and a bound that treats them as requirements must pay for each one.

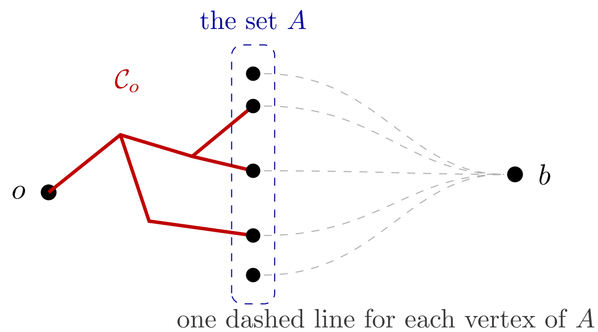


Figure 5: Connecting through a set. The source on the left reaches the set in the middle with high probability, and each vertex of that set reaches the target on the right with high probability. The union bound pays once for every dashed line, so it is useless when the set is large; the gluing inequalities of Section 3.2 pay only once in total.

The size of the set cannot be controlled in advance, which is why a new inequality is needed rather than a sharper application of an old one. In the reduction of Section 3.3, the set is the set of vertices through which a crossing of one block can enter the next, and its size grows with the block scale. There is no bound on it available in advance.

3.2 Three gluing inequalities

Three statements must be kept apart. Each is an assertion about every finite family of independent hyperedges, every o and b, and every nonempty finite set A; that quantifier is not repeated below. They differ in how the loss accumulates. The three names are used only in these notes, and Kozma and Nitzan [27] state the first and the third as their Conjectures 1 and 3.

Definition 3.2 (Multiplicative Gluing) **Multiplicative gluing** is the assertion that

$$\mathbb{P}(o \leftrightarrow b) \geq \mathbb{P}(o \leftrightarrow A) \cdot \min_{a \in A} \mathbb{P}(a \leftrightarrow b)\,.$$

The loss is a factor, so it compounds when the inequality is applied repeatedly.

Definition 3.3 (Additive Gluing) **Additive gluing** is the assertion that for every $t \geq 0$, if $\mathbb{P}(a \leftrightarrow b) \geq 1 - t$ for every $a \in A$, then

$$\mathbb{P}(o \leftrightarrow b) \geq \mathbb{P}(o \leftrightarrow A) - t\,.$$

Here the loss is a summand, and it does not depend on the size of A.

Definition 3.4 (Near-One Gluing) **Near-one gluing** is the assertion that for every $\varepsilon > 0$ there is $\delta > 0$, depending on ε alone, such that if $\mathbb{P}(o \leftrightarrow A) > 1 - \delta$ and $\mathbb{P}(a \leftrightarrow b) > 1 - \delta$ for every $a \in A$, then $\mathbb{P}(o \leftrightarrow b) > 1 - \varepsilon$.

The three are listed in decreasing order of strength, and the implications are easy.

Lemma 3.5. (Multiplicative Gluing Implies Additive Gluing) *If multiplicative gluing holds, then additive gluing holds.*

Proof. Assume multiplicative gluing, and let $t \geq 0$ with $\mathbb{P}(a \leftrightarrow b) \geq 1 - t$ for every $a \in A$. If $t \geq 1$ the conclusion is immediate because $\mathbb{P}(o \leftrightarrow b) \geq 0 \geq \mathbb{P}(o \leftrightarrow A) - t$. So assume $t < 1$. Then

$$\mathbb{P}(o \leftrightarrow b) \geq \mathbb{P}(o \leftrightarrow A)(1-t) = \mathbb{P}(o \leftrightarrow A) - t\,\mathbb{P}(o \leftrightarrow A) \geq \mathbb{P}(o \leftrightarrow A) - t\,,$$

using $\mathbb{P}(o \leftrightarrow A) \leq 1$ in the last step. □

Lemma 3.6. (Additive Gluing Implies Near-One Gluing) *If additive gluing holds, then near-one gluing holds, with $\delta = \varepsilon/2$.*

Proof. Let $\varepsilon > 0$ and set $\delta = \varepsilon/2$. Suppose $\mathbb{P}(o \leftrightarrow A) > 1 - \delta$ and $\mathbb{P}(a \leftrightarrow b) > 1 - \delta$ for every $a \in A$. Applying additive gluing with $t = \delta$,

$$\mathbb{P}(o \leftrightarrow b) \geq \mathbb{P}(o \leftrightarrow A) - \delta > 1 - 2\delta = 1 - \varepsilon\,.$$

The number δ depends only on ε, as required. □

Sections 4 and 5 prove the multiplicative statement, and Lemmas 3.5 and 3.6 then give the other two. The near-one form is what the next section consumes.

3.3 The reduction to a finite statement

The following theorem is the reason the gluing inequalities are worth proving. It is Theorem 6 of Kozma and Nitzan [27], on page 15 of that paper.

Theorem 3.7. (Near-One Gluing Implies the Percolation Probability Vanishes at Criticality) *Suppose near-one gluing holds. Then for every integer $d \geq 2$ the percolation probability of Bernoulli bond percolation on $\mathbb{Z}^d$ satisfies $\theta(p_c(d)) = 0$.*

Nothing below assumes Theorem 3.7. It is a theorem of Kozma and Nitzan [27], and it is proved in the Lean development that accompanies these notes; we cite it rather than reproduce it, and its shape matters for reading the rest, so we describe the two cases.

In two dimensions the conclusion is classical and the gluing hypothesis is not needed: the critical probability equals 1/2 and the percolation probability vanishes there, by Harris [23] and Kesten [24].

For $d \geq 3$ the argument is a renormalization. Suppose for a contradiction that $\theta(p_c) > 0$. Partition $\mathbb{Z}^d$ into blocks of a large side length and declare a block good when the configuration inside it contains a crossing cluster joining all its faces. Percolation of good blocks would give an infinite open cluster, so the aim is to show that blocks are good with probability close enough to 1. What has to be checked is that the crossing cluster of one block connects to the crossing cluster of the next, and this is exactly the problem of connecting through a set: the source is the crossing cluster of the first block, the set is the set of vertices on the shared face through which the connection can pass, and the target is the crossing cluster of the second block. The set is the face of a block, so its size grows with the block scale, which is precisely why the union bound of Lemma 3.1 cannot be used and why near-one gluing is the right hypothesis. The block percolation so obtained yields percolation in a slab, and the strict inequality for critical probabilities under an essential enhancement, proved by Aizenman and Grimmett [4], then forces $p > p_c$, which is the contradiction. One-step renormalization of this kind originates with Grimmett and Marstrand [20]; the absence of percolation at the critical

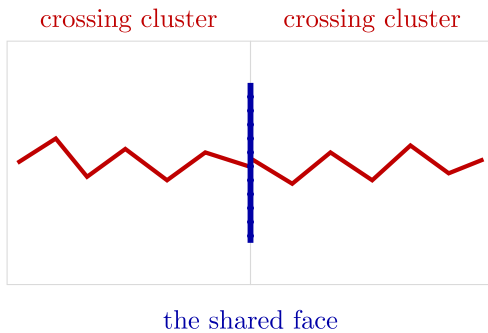


Figure 6: The set in the renormalization of Theorem 3.7. The crossing cluster of the left block must be joined to the crossing cluster of the right block, and the connection may cross the shared face, drawn in bold, at a vertex that is not fixed in advance. Since the number of such vertices grows with the side length of the blocks, the union bound over them is unusable.

probability in half-spaces is due to Barsky, Grimmett, and Newman [5], and the corresponding statement inside a slab to Duminil-Copin, Sidoravicius, and Tassion [13]. A textbook account of block renormalization is Grimmett [19, §7.1–7.2].

Two features of Theorem 3.7 deserve emphasis. The hypothesis is a statement about arbitrary finite families of independent hyperedges, so it carries no geometry: nothing in near-one gluing knows about dimension, about the lattice, or about translation invariance. And the number δ must not depend on the set, which is what allows the block scale to be chosen after δ is fixed.

3.4 What remains to be proved

Combining the results of this section reduces the lattice problem to a single finite inequality.

Corollary 3.8. (Additive Gluing Implies the Percolation Probability Vanishes at Criticality) *Suppose additive gluing holds. Then for every integer $d \geq 2$ the percolation probability of Bernoulli bond percolation on $\mathbb{Z}^d$ satisfies $\theta(p_c(d)) = 0$.*

Proof. Additive gluing implies near-one gluing by Lemma 3.6, and near-one gluing implies the conclusion by Theorem 3.7. □

By Theorem 1.15 the conclusion of Corollary 3.8 is equivalent to the continuity of the percolation probability on the whole parameter interval, so proving additive gluing settles the continuity question in every dimension.

Everything therefore rests on the gluing inequalities, and the two sections that follow prove them. The route is indirect. Section 4 constructs a family of inequalities comparing conditioned covariances at different vertices, and Section 5 converts that family into additive gluing by an induction on the size of the set, charging the loss not to every vertex of the set but only to one of them.

4 A Family of Conditioned Covariance Inequalities

Section 3 reduced the vanishing of the percolation probability at the critical probability to one finite inequality. This section builds the object that inequality is derived from: a family of inequalities comparing conditioned covariances at two vertices. It is due to Anthropic, as is the derivation in Section 5, and it does not appear in the earlier literature.

The result to reach is Theorem 4.9, in Section 4.4. It says that a certain alternating sum of conditioned covariances is nonnegative. The data are a finite family of independent hyperedges, a vertex x, a set Y that the cluster of x must avoid, two further vertices o and v, and a finite list of further vertices $d_1, \ldots, d_\ell$. Take Y empty, the list empty, and f the indicator that x is connected to v, and what is left is the Harris inequality of Section 2. So the theorem extends positive correlation in two directions at once: it survives conditioning, and it survives subtracting off the contributions of other vertices one at a time. The second direction is what the induction in Section 5 needs, and it is why the vertices $d_1, \ldots, d_\ell$ are carried at all.

Section 4.1 introduces the conditioned covariance and records what Section 2 already gives about it, namely that it is nonnegative. Section 4.2 states the comparison between o and v, which is the first level of the family, and checks that in the simplest case it is exactly the Harris inequality. Section 4.3 introduces the vertices $d_1, \ldots, d_\ell$ and the alternating sum, and proves the algebraic identity the induction turns on. Section 4.4 states the theorem. Section 4.5 records a consequence used in Section 5: for fixed data the whole family reduces to finitely many inequalities in the edge parameters.

Throughout this section the model is the one of Definition 2.7, x is a vertex, and Y is a set of vertices with $x \notin Y$. Every conditioning is on the event $\{x \not\leftrightarrow Y\}$ that the cluster of x reaches no vertex of Y, and that event is assumed to have positive probability. For bond percolation the set $\mathcal{K}_x$ of Definition 2.8 is the set of open edges with an endpoint in $\mathcal{C}_x$.

4.1 The conditioned covariance

Definition 4.1 (Conditioned Covariance) Let f be an increasing real function of $\mathcal{K}_x$. For a vertex u define

$$\kappa(u) := \operatorname{Cov}\big(f(\mathcal{K}_x),\, \mathbf{1}\{u \in \mathcal{C}_x\} \,\big|\, \{x \not\leftrightarrow Y\}\big)\,. \tag{23}$$

Both arguments of the covariance are increasing functions of $\mathcal{K}_x$, so Corollary 2.14 applies and gives the following at once.

Lemma 4.2. (The Conditioned Covariance Is Nonnegative) *In the setting of Definition 4.1, $\kappa(u) \geq 0$ for every vertex u, and $\kappa(u) = 0$ whenever $u \in Y$.*

Proof. Both $f(\mathcal{K}_x)$ and $\mathbf{1}\{u \in \mathcal{C}_x\}$ are increasing functions of $\mathcal{K}_x$, so their conditioned covariance is nonnegative by Corollary 2.14.

If $u \in Y$ then on the event $\{x \not\leftrightarrow Y\}$ the cluster of x does not reach u, so the indicator $\mathbf{1}\{u \in \mathcal{C}_x\}$ vanishes identically under the conditioned measure, and a covariance with a constant is zero. □

Lemma 4.2 is all that classical correlation theory gives, and it is not enough. What the argument of Section 5 needs is a comparison between the values of κ at two different vertices, and no amount of positivity supplies one.

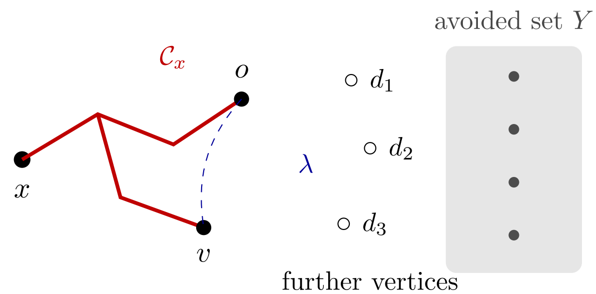


Figure 7: The data of Theorem 4.9. The cluster of x is forbidden to reach the shaded set Y, drawn as the shaded region on the right. The vertices o and v are compared: the value of the conditioned covariance at the first is bounded below by the value at the second, discounted by the number λ. The vertices $d_1, \ldots, d_\ell$ of Section 4.3 are the open circles, and the constant attached to d_j is computed with $d_1, \ldots, d_{j-1}$ already avoided.

4.2 Comparing the vertices o and v

Fix two distinct vertices o and v, neither in Y and neither equal to x. The comparison we need transports the value of κ at v to its value at o, at a price given by the probability that the two vertices o and v are joined to each other.

Definition 4.3 (The Number Lambda) Let o and v be distinct vertices outside $Y \cup \{x\}$, and let $d_1, \ldots, d_\ell$ be finitely many further vertices, distinct from one another and from x, o and v, and none of them in Y. Assume the event that v avoids $Y \cup \{x\} \cup \{d_1, \ldots, d_\ell\}$ has positive probability. The number λ is

$$\lambda := \mathbb{P}\big(o \in \mathcal{C}_v \,\big|\, \{v \nleftrightarrow Y \cup \{x\} \cup \{d_1, \ldots, d_\ell\}\}\big)\,, \tag{24}$$

the probability that o lies in the open cluster of v, given that v reaches none of x, of Y, and of $d_1, \ldots, d_\ell$. When there are no vertices d_j the conditioning is on v avoiding $Y \cup \{x\}$ alone.

The conditioning in (24) must include every d_j. Omitting them gives a larger number, and the comparison below then fails: on the star with center d_1 and leaves x, o, v, all three edges open with probability $1/2$, taking Y empty and $f(\mathcal{K}_x) = \mathbf{1}\{v \in \mathcal{C}_x\}$, the inequality with the smaller conditioning set would read $0 \geq 1/48$.

Two features of (24) matter. The conditioning is on an avoidance event for v, not for x, so λ is computed under a different conditioning from the one defining κ. And λ does not depend on the function f.

Definition 4.4 (The Comparison at Level Zero) In the setting above, the comparison at level zero is the assertion that

$$\kappa(o) \geq \lambda\, \kappa(v) \tag{25}$$

holds for every increasing real function f of $\mathcal{K}_x$.

In its simplest case (25) reduces to a statement already available.

Lemma 4.5. (Level Zero Contains the Harris Inequality) *Let x, o, v be distinct vertices, take Y empty, and take $f(\mathcal{K}_x) = \mathbf{1}\{v \in \mathcal{C}_x\}$. Then the comparison at level zero (25) is equivalent* *to the inequality*

$$\mathbb{P}\big(o \leftrightarrow \{x,v\} \text{ and } x \leftrightarrow v\big) \geq \mathbb{P}\big(o \leftrightarrow \{x,v\}\big)\,\mathbb{P}\big(x \leftrightarrow v\big)\,, \tag{26}$$

which is an instance of the Harris inequality, Theorem 2.3.

Proof. With Y empty there is no conditioning, and with $f(\mathcal{K}_x) = \mathbf{1}\{v \in \mathcal{C}_x\}$ the covariance (23) becomes the unconditioned covariance of the two indicators $\mathbf{1}\{v \in \mathcal{C}_x\}$ and $\mathbf{1}\{o \in \mathcal{C}_x\}$. Both are increasing events, namely $\{x \leftrightarrow v\}$ and $\{x \leftrightarrow o\}$, so (25) compares two covariances of pairs of increasing events. Expanding the covariances and collecting the terms in which the two clusters meet gives (26), and the events $\{o \leftrightarrow \{x,v\}\}$ and $\{x \leftrightarrow v\}$ are both increasing, so Theorem 2.3 applies. □

The general case of (25), with Y nonempty and f an arbitrary increasing function, is much stronger, and it is not a consequence of the Harris inequality: the conditioning destroys the product structure that the proof of Theorem 2.3 consumed, as Example 2.6 showed.

4.3 Further vertices, and peeling

The induction of Section 5 removes vertices from a set one at a time, and a removed vertex does not leave the argument: it stays as a vertex whose contribution must be subtracted. The family therefore has to carry a list of such vertices from the start.

Definition 4.6 (The Vertices d_j and Their Constants) Let $d_1, \ldots, d_\ell$ be distinct vertices, none equal to x, to o or to v, and none in Y. For $1 \leq j \leq \ell$ define

$$c_j(u) := \mathbb{P}\big(u \in \mathcal{C}_{d_j} \,\big|\, \{d_j \not\leftrightarrow Y \cup \{x\} \cup \{d_1, \ldots, d_{j-1}\}\}\big)\,, \tag{27}$$

the probability that u lies in the open cluster of d_j, given that d_j reaches neither x, nor Y, nor an earlier vertex of the list; the conditioning event is assumed to have positive probability. Every hyperedge probability strictly between 0 and 1 makes all of these events have positive probability, which is the situation of Theorem 4.9.

The set avoided in (27) grows along the list, so the constants are computed in a definite order and c_j depends on $d_1, \ldots, d_{j-1}$. This ordering is not a convenience; it is what makes the induction close.

Given these vertices and their constants we peel one vertex at a time. The operation applies to an arbitrary real function of the vertices, not only to κ, and that generality is needed in Section 5.

Definition 4.7 (Peeling, and the Functional M) Let $c_1, \ldots, c_\ell$ be as in (27) and let λ be as in (24). For a real function q of the vertices set $q^{[0]} = q$ and

$$q^{[j]}(u) = q^{[j-1]}(u) - c_j(u)\, q^{[j-1]}(d_j) \qquad (1 \leq j \leq \ell)\,, \tag{28}$$

and put

$$M[q] = q^{[\ell]}(o) - \lambda\, q^{[\ell]}(v)\,. \tag{29}$$

The j-th step subtracts from the value at u the value at d_j, weighted by the conditional connection probability $c_j(u)$. Both operations are linear.

Lemma 4.8. (Peeling and M Are Linear) *For each j with $0 \le j \le \ell$ the map $q \mapsto q^{[j]}$ is linear, and $q \mapsto M[q]$ is linear.*

Proof. We induct on j. For $j = 0$ the map is the identity. For the step, $q^{[j]}$ is obtained from $q^{[j-1]}$ by subtracting the number $q^{[j-1]}(d_j)$ times the fixed function c_j, and $c_1, \ldots, c_\ell$ do not depend on q; a difference of linear maps is linear. The functional M is a difference of two evaluations of $q^{[\ell]}$, so it is linear as well. □

4.4 The family of inequalities

We can now state the main result of this section. Recall that κ depends on the increasing function f through (23).

Theorem 4.9. (The Family of Inequalities) *Let every hyperedge probability lie strictly between 0 and 1. Let x be a vertex, let Y be a set of vertices with $x \notin Y$, let o and v be distinct vertices outside $Y \cup \{x\}$, and let $d_1, \ldots, d_\ell$ be distinct vertices, none equal to x, to o or to v, and none in Y. Then for every increasing real function f of $\mathcal{K}_x$,*

$$M[\kappa] \;\ge\; 0\,. \tag{30}$$

Written out, the case $\ell = 0$ of (30) is the level zero comparison (25), and the case $\ell = 1$ reads

$$\kappa(o) - c_1(o)\,\kappa(d_1) \;\ge\; \lambda\big(\kappa(v) - c_1(v)\,\kappa(d_1)\big)\,.$$

Each successive level compares the same two vertices after removing more of the contribution of $d_1, \ldots, d_\ell$.

The proof below is due to Anthropic. It uses the six results of Section 2.6 and nothing else that has not already been proved.

Proof. Since every hyperedge probability is less than one, isolating each named vertex has positive probability. Thus every conditional probability used below has a positive denominator. Adding a constant to f changes no covariance, so we may and do suppose that f is nonnegative.

For a vertex function h, write

$$(L_j h)(u) = h(u) - c_j(u)h(d_j).$$

Then $h^{[j]} = L_j \cdots L_1 h$. We prove the theorem by strong induction on ℓ. The argument below includes the case $\ell = 0$, for which every sum indexed by j is empty.

Step 1: remove the conditioning cluster. Let

$$\nu = \mathbb{P}(\,\cdot \mid x \nleftrightarrow Y).$$

Under ν, expose $\mathcal{K}_Y$ and put

$$W = U \setminus C_Y.$$

This set contains x. Lemma 2.12 says that, conditional on what was exposed, the hyperedges of the model induced on W have their original independent laws. Let π be the law of W under ν. For an increasing function g of $\mathcal{K}_x$, define

$$\kappa_g^W(u) = \mathrm{Cov}_W\big(g(\mathcal{K}_x^W), \mathbf{1}\{u \in C_x^W\}\big)$$

and

$$\Delta_\ell(g) = \mathbb{E}_\pi\left[(L_\ell \cdots L_1 \kappa_g^W)(o) - \lambda(L_\ell \cdots L_1 \kappa_g^W)(v)\right]. \tag{31}$$

The constants c_j and λ in this formula are those of the theorem.

We next show that it is enough to prove

$$\Delta_\ell(g) \geq 0 \tag{32}$$

for every nonnegative increasing g. By linearity there are constants β_u, indexed by o, v and $d_1, \ldots, d_\ell$, such that

$$M[h] = \sum_u \beta_u h(u).$$

For a set A of hyperedges, set

$$H(A) = \sum_u \beta_u \mathbf{1}\{u \in C_x(A)\},$$

where $C_x(A)$ is the component of x in the open hyperedges of A. If

$$\mathfrak{M}_\ell(g) = M[\kappa_g],$$

then

$$\mathfrak{M}_\ell(g) = \operatorname{Cov}_\nu(g(\mathcal{K}_x), H(\mathcal{K}_x)). \tag{33}$$

Write $A = \mathcal{K}_x$ and $B = \mathcal{K}_Y$ under ν. Let P_A and P_B denote conditional expectation given A and B, respectively, and define the two-step resampling operator

$$\mathcal{A} = P_A P_B, \qquad (\mathcal{A}g)(A) = \mathbb{E}_\nu\left[\mathbb{E}_\nu[g(A) \mid B] \mid A\right].$$

The law of total covariance, first conditional on B, gives

$$\mathfrak{M}_\ell(g) = \Delta_\ell(g) + \mathfrak{M}_\ell(\mathcal{A}g). \tag{34}$$

Here is the calculation. Put $h(B) = \mathbb{E}_\nu[g(A) \mid B]$. The first term in the law of total covariance is

$$\mathbb{E}_\nu[\operatorname{Cov}_\nu(g(A), H(A) \mid B)] = \Delta_\ell(g)$$

by Lemma 2.12. Since $H(A)$ is measurable with respect to A, the second term is

$$\begin{aligned}\operatorname{Cov}_\nu(h(B), \mathbb{E}_\nu[H(A) \mid B]) &= \operatorname{Cov}_\nu(h(B), H(A)) \\ &= \operatorname{Cov}_\nu(\mathbb{E}_\nu[h(B) \mid A], H(A)) = \mathfrak{M}_\ell(\mathcal{A}g).\end{aligned}$$

We record two properties of $\mathcal{A}$. Given $B = J$, exact cluster conditioning makes the x-cluster an independent cluster in the graph obtained by deleting the support of J. Hence

$$J \longmapsto \mathbb{E}_\nu[g(A) \mid B = J]$$

is decreasing when g is increasing. Given $A = I$, the Y-cluster has the product law off the support of I. If $I \subseteq I'$, couple the two laws using the same hyperedge states. The set $\mathcal{K}_Y$

obtained after deleting the larger support is contained in the other one. Applying the preceding decreasing conditional mean shows

$$(\mathcal{A}g)(I) \leq (\mathcal{A}g)(I').$$

Thus $\mathcal{A}$ preserves increasing functions. A function defined only on the values it can take is extended, when needed, by

$$\overline{g}(J) = \max\bigl(\{g(I) : I \subseteq J \text{ and } g \text{ is defined at } I\} \cup \{0\}\bigr).$$

This agrees with g wherever g was defined and is increasing on the entire Boolean lattice. The operator $\mathcal{A}$ also preserves nonnegativity and constants.

There is a number $\epsilon > 0$ such that the transition represented by $\mathcal{A}$ has a fixed component of mass at least ϵ. Indeed, during the resampling of B, close every still available edge incident to Y. This has probability at least

$$\epsilon = \prod_{e:\, \iota(e)\cap Y \neq \varnothing} (1 - p_e) > 0$$

and forces $C_Y = Y$. The subsequent resampling of A then has a fixed law, independent of its previous value. Consequently

$$\operatorname{osc}(\mathcal{A}^n g) \leq (1-\epsilon)^n \operatorname{osc}(g). \tag{35}$$

If $Y = \varnothing$, the first application of $\mathcal{A}$ is already constant. Otherwise, every averaging operator raises the minimum and lowers the maximum. Equation (35) shows that these bounds approach the same number, so $\mathcal{A}^n g$ converges uniformly to a constant.

Iterating (34) gives

$$\mathfrak{M}_\ell(f) = \sum_{i=0}^{n-1} \Delta_\ell(\mathcal{A}^i f) + \mathfrak{M}_\ell(\mathcal{A}^n f).$$

The last term tends to zero because $\mathfrak{M}_\ell$ is a linear functional on a finite state space and vanishes on constants. Therefore (32) for nonnegative increasing functions implies $\mathfrak{M}_\ell(f) \geq 0$.

Step 2: add the vertices d_j one at a time. Fix a nonnegative increasing function g, and put

$$S_j = \{x, d_1, \ldots, d_j\}, \qquad S_0 = \{x\}, \qquad X = S_\ell.$$

For $T \subseteq U$ and a vertex u, define

$$r_T^W(u) = \operatorname{Cov}_W\bigl(g(\mathcal{K}_x^W), \mathbf{1}\{u \leftrightarrow T\}\bigr), \qquad R_j(u) = \mathbb{E}_\pi[r_{S_j}^W(u)].$$

Thus

$$R_0(u) = \mathbb{E}_\pi[\kappa_g^W(u)]$$

and, on putting

$$h_u(W) = \operatorname{Cov}_W\bigl(g(\mathcal{K}_x^W), \mathbf{1}\{u \leftrightarrow X\}\bigr),$$

we have $R_\ell(u) = \mathbb{E}_\pi[h_u(W)]$.

Use the deletion notation of Lemma 2.20 and define

$$\Phi(K) = \mathbb{E}\left[\mathbf{1}\{C_x^{-K} \cap Y = \varnothing\}\left(g(\mathcal{K}_x^{-C_Y^{-K}}) - g(\mathcal{K}_x^{-K})\right)\right]. \tag{36}$$

We need the fact that Φ is nonnegative and increasing on sets that avoid Y. This follows pointwise. For a fixed configuration, write

$$A_K = \{C_x^{-K} \cap Y = \varnothing\}, \qquad U_K = \mathcal{K}_x^{-C_Y^{-K}}, \qquad V_K = \mathcal{K}_x^{-K}.$$

On A_K, the x-cluster and the Y-cluster in the graph with K deleted are disjoint. Restoring the edges meeting K and then deleting the edges meeting C_Y^{-K} cannot remove an edge of the former x-cluster. Thus

$$A_K \quad \Longrightarrow \quad V_K \subseteq U_K. \tag{37}$$

The integrand in (36) is therefore nonnegative.

Now let $K \subseteq K'$ and suppose $K' \cap Y = \varnothing$. Deleting more vertices gives

$$C_Y^{-K'} \subseteq C_Y^{-K}, \qquad V_{K'} \subseteq V_K, \qquad U_K \subseteq U_{K'},$$

and $A_K \subseteq A_{K'}$. On A_K, monotonicity of g gives

$$g(U_K) - g(V_K) \leq g(U_{K'}) - g(V_{K'}).$$

On $A_{K'} \setminus A_K$, the expression on the right is nonnegative by (37). Hence $\Phi(K) \leq \Phi(K')$. Extend Φ to all subsets of U by

$$\overline{\Phi}(K) = \max\{\Phi(I) : I \subseteq K,\ I \cap Y = \varnothing\}. \tag{38}$$

The extension is nonnegative and increasing, and it agrees with Φ whenever $K \cap Y = \varnothing$. Every cluster to which it is applied below avoids Y, so we write Φ for the extension.

For $1 \leq j \leq \ell$, set

$$Y_j = Y \cup \{x, d_1, \ldots, d_{j-1}\}, \qquad \alpha_j = \frac{\mathbb{P}(d_j \nleftrightarrow Y_j)}{\mathbb{P}(x \nleftrightarrow Y)}$$

and define

$$\eta_j(u) = \operatorname{Cov}\left(\Phi(C_{d_j}), \mathbf{1}\{u \in C_{d_j}\} \,\middle|\, d_j \nleftrightarrow Y_j\right).$$

Apply Lemma 2.20 with $z = d_j$ and $S = S_{j-1}$. Its constant $c(u)$ is precisely $c_j(u)$, and it gives

$$L_j R_{j-1} - R_j = \alpha_j \eta_j. \tag{39}$$

Set

$$D_j = L_j \cdots L_1 R_0 - R_j, \qquad D_0 = 0.$$

Equation (39) gives

$$D_j = L_j D_{j-1} + \alpha_j \eta_j.$$

Iteration yields

$$D_\ell = \sum_{j=1}^{\ell} \alpha_j L_\ell L_{\ell-1} \cdots L_{j+1} \eta_j, \tag{40}$$

where the operator following α_ℓ is the identity. Since the operators L_j commute with expectation,

$$\begin{aligned} \Delta_\ell(g) = {} & \mathbb{E}_\pi[h_o(W) - \lambda h_v(W)] \\ & + \sum_{j=1}^{\ell} \alpha_j \left[(L_\ell \cdots L_{j+1}\eta_j)(o) - \lambda (L_\ell \cdots L_{j+1}\eta_j)(v)\right]. \end{aligned} \tag{41}$$

The expression in square brackets is an instance of the theorem with $\ell - j$ vertices. Its distinguished vertex is d_j, its avoided set is Y_j, and its ordered list is $d_{j+1}, \ldots, d_\ell$. The constants in its successive operators are the original $c_{j+1}, \ldots, c_\ell$, because

$$Y_j \cup \{d_j, d_{j+1}, \ldots, d_{\ell-1}\} = Y \cup \{x, d_1, \ldots, d_{\ell-1}\}.$$

Its final connection constant is λ, since its final avoided set is

$$Y_j \cup \{d_j, d_{j+1}, \ldots, d_\ell\} = Y \cup X.$$

The function $\Phi(C_{d_j})$ is a nonnegative increasing function of the open hyperedges of the cluster of d_j. The strong induction hypothesis therefore says that every term in the sum in (41) is nonnegative.

Step 3: compare o and v in each induced hypergraph. For a fixed W, define

$$q(W) = \begin{cases} \mathbb{P}_W(o \leftrightarrow v \mid v \not\leftrightarrow X), & v \in W, \\ 0, & v \notin W. \end{cases}$$

We first prove

$$h_o(W) \geq q(W) h_v(W). \tag{42}$$

If $v \notin W$, then $q(W) = h_v(W) = 0$ and the assertion is the Harris inequality for $h_o(W)$. If $o \notin W$, both $q(W)$ and $h_o(W)$ are zero. We may therefore suppose $o, v \in W$.

Put

$$m = \mathbb{E}_W[g(\mathcal{K}_x^W)], \qquad D = \{v \not\leftrightarrow X\}, \qquad J = \{o \leftrightarrow v,\ v \not\leftrightarrow X\}.$$

The disjoint union

$$\{o \leftrightarrow X \cup \{v\}\} = \{o \leftrightarrow X\} \,\dot\cup\, J$$

and the Harris inequality give

$$h_o(W) \geq -\mathbb{E}_W[(g(\mathcal{K}_x^W) - m); J]. \tag{43}$$

Conditional on D, apply Lemma 2.15 to the source sets X and $\{v\}$. The function $g(\mathcal{K}_x^W)$ is increasing in $\mathcal{K}_X^W$, and $1 - \mathbf{1}\{o \in C_v^W\}$ is decreasing in $\mathcal{K}_v^W$. Their positive correlation is equivalent to

$$\mathbb{P}_W(D)\mathbb{E}_W[(g(\mathcal{K}_x^W) - m); J] \leq \mathbb{P}_W(J)\mathbb{E}_W[(g(\mathcal{K}_x^W) - m); D].$$

Since

$$h_v(W) = -\mathbb{E}_W[(g(\mathcal{K}_x^W) - m); D], \qquad q(W) = \frac{\mathbb{P}_W(J)}{\mathbb{P}_W(D)},$$

this inequality and (43) prove (42).

It remains to compare the varying number $q(W)$ with λ. Apply Lemma 2.18 with $A = X$ and $F(W) = h_v(W)$. The Harris inequality gives $F(W) \geq 0$, and $F(W) = 0$ if $v \notin W$. Lemma 2.17, with source $S = X$ and cluster function g, is exactly the assumption (11).

The conclusion (12) gives

$$\begin{aligned} &\mathbb{P}(o \leftrightarrow v,\ v \not\leftrightarrow X \cup Y)\, \mathbb{E}[h_v(U \setminus C_Y); x \notin C_Y] \\ &\qquad \leq \mathbb{P}(v \not\leftrightarrow X \cup Y)\, \mathbb{E}[q(U \setminus C_Y) h_v(U \setminus C_Y); x \notin C_Y]. \end{aligned}$$

The first probability is λ times the second. The second is positive because v can be isolated, so cancellation gives

$$\mathbb{E}[q(W)h_v(W); x \nleftrightarrow Y] \geq \lambda \mathbb{E}[h_v(W); x \nleftrightarrow Y]. \tag{44}$$

Divide by $\mathbb{P}(x \nleftrightarrow Y)$ and average (42). Equation (44) then shows that

$$\mathbb{E}_\pi[h_o(W) - \lambda h_v(W)] \geq 0.$$

This proves that the first term in (41) is nonnegative. Step 2 and the induction hypothesis show that all its other terms are nonnegative as well. Hence $\Delta_\ell(g) \geq 0$ for every nonnegative increasing g. Step 1 now gives $M[\kappa] = \mathfrak{M}_\ell(f) \geq 0$, as required. □

4.5 Reduction to indicator functions

One structural consequence of Theorem 4.9 is used repeatedly in Section 5, and it follows from linearity alone.

Proposition 4.10. (Linearity in f) *Fix the model, the vertices x, o, v, the set Y and the list $d_1, \ldots, d_\ell$. The quantity $M[\kappa]$ is a linear function of f, and it vanishes when f is constant.*

Proof. The map sending f to κ is linear, because a covariance is linear in each of its arguments and the second argument of (23) does not involve f. The map sending κ to $M[\kappa]$ is linear by Lemma 4.8, since λ and the constants c_j do not depend on f. A composition of linear maps is linear.

If f is constant then $f(\mathcal{K}_x)$ is a constant random variable, so every covariance (23) vanishes, hence κ is identically zero and so is the displayed quantity. □

A consequence of Proposition 4.10 is that it suffices to verify (30) when f is the indicator of an increasing family of sets of hyperedges, since every increasing function on a finite family of sets is a nonnegative combination of such indicators together with a constant. For fixed data Theorem 4.9 is therefore a finite family of inequalities, one for each such indicator. The quantities appearing in them are rational functions of the hyperedge probabilities rather than polynomials, since λ and the constants c_j are ratios of two probabilities; clearing the positive denominators turns each inequality into a polynomial one.

5 Deriving the Gluing Inequalities

Section 4 produced a family of inequalities comparing conditioned covariances at two vertices. This section converts that family into the gluing inequality that Section 3 showed to be enough. Everything here is due to Anthropic.

The conversion turns on one idea. When a source reaches a set of vertices, charge the comparison to one vertex of that set and to no other: the vertex of least rank among those the source meets. Each configuration in which the source reaches the set names exactly one such vertex, so nothing is summed over the set, and the estimate does not degrade as the set grows. Degrading as the set grows is exactly how the union bound of Lemma 3.1 fails. What the argument produces directly is the multiplicative form, and the other two follow from it.

Section 5.1 introduces the vertex of least rank and the quantity measured against it. Section 5.2 states the inequality that moves that quantity from one vertex to another. Section 5.3

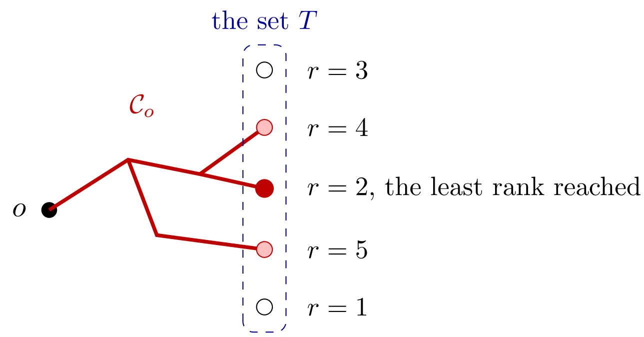


Figure 8: The vertex of least rank. The cluster of o reaches three vertices of T, drawn filled, and misses two, drawn open. Of the three it reaches, the one of least rank is the vertex of rank 2, and the whole comparison is charged to that vertex and to no other. The vertex of rank 1 is not reached, so it plays no part.

iterates that inequality over the set, and says why the auxiliary vertices of Section 4 are needed for the iteration to close. Section 5.4 completes the derivation, and Section 5.5 draws the consequence for the lattice, which is the theorem these notes set out to prove.

5.1 The vertex of least rank and the quantity $D_u(T)$

Throughout this section the model is the one of Definition 2.7, Y is a set of vertices, F is an increasing real function of sets of vertices, and T is a finite set of vertices disjoint from Y with $\mathbb{P}(\{x \nleftrightarrow Y\}) > 0$ for every $x \in T$. Carrying the set Y costs nothing and is what makes the induction below close. For $x \in T$ write

$$m_x = \frac{\mathbb{E}\big[F(\mathcal{C}_x)\,;\,\{x \nleftrightarrow Y\}\big]}{\mathbb{P}(\{x \nleftrightarrow Y\})} \tag{45}$$

for the conditional expectation of $F(\mathcal{C}_x)$ given that x avoids Y.

Definition 5.1 (Rank Compatible with the Means) A function r from the vertices to the real numbers is a **compatible rank** for T when r is injective on T and $m_x \le m_y$ for all $x, y \in T$ with $r(x) < r(y)$.

A compatible rank exists: order T by the values m_x and break ties arbitrarily.

Definition 5.2 (The Events J_x and the Quantity $D_u(T)$) Let r be a compatible rank and let u be a vertex. For $x \in T$ put

$$J_x = \{u \nleftrightarrow Y\} \cap \{u \leftrightarrow x\} \cap \bigcap_{y \in T,\, r(y) < r(x)} \{u \nleftrightarrow y\}\,, \tag{46}$$

the event that u avoids Y and that x is the vertex of smallest rank in $T \cap \mathcal{C}_u$, and set

$$D_u(T) = \mathbb{E}\big[F(\mathcal{C}_u)\,;\,\{u \nleftrightarrow Y\},\ u \leftrightarrow T\big] - \sum_{x \in T} \mathbb{P}(J_x)\, m_x\,. \tag{47}$$

Lemma 5.3. (The Events J_x Partition, and $D_u(T)$ Does Not Depend on the Rank) *The events J_x for $x \in T$ are pairwise disjoint and their union is $\{u \not\leftrightarrow Y\} \cap \{u \leftrightarrow T\}$. Moreover*

$$D_u(T) = \mathbb{E}\Big[\mathbf{1}\{\{u \not\leftrightarrow Y\},\ u \leftrightarrow T\}\Big(F(\mathcal{C}_u) - \min_{x \in T \cap \mathcal{C}_u} m_x\Big)\Big], \tag{48}$$

so $D_u(T)$ does not depend on the choice of compatible rank.

Proof. If J_x and J_y both occur with $x \neq y$, we may assume $r(x) < r(y)$ since r is injective on T; then J_y asserts $u \not\leftrightarrow x$ while J_x asserts $u \leftrightarrow x$, a contradiction. If u avoids Y and reaches T, then $T \cap \mathcal{C}_u$ is nonempty and finite, so it has a unique element of smallest rank, and J_x occurs for that element. Conversely each J_x is contained in $\{u \not\leftrightarrow Y\} \cap \{u \leftrightarrow T\}$.

For (48), note that on J_x the vertex x has the smallest rank among the vertices of T in $\mathcal{C}_u$, hence by compatibility the smallest value of m, so $m_x = \min_{y \in T \cap \mathcal{C}_u} m_y$ there. Summing over the partition turns $\sum_x \mathbb{P}(J_x) m_x$ into the expectation of that minimum over $\{u \not\leftrightarrow Y\} \cap \{u \leftrightarrow T\}$, which gives (48). The right side of (48) does not mention r. $\square$

Two cases are immediate and are used below. If $u \in Y$ then $\{u \not\leftrightarrow Y\}$ is empty and $D_u(T) = 0$. If $u \in T$ then $u \in T \cap \mathcal{C}_u$ on $\{u \not\leftrightarrow Y\}$, so the minimum in (48) is at most m_u, whence $D_u(T) \geq \mathbb{E}[F(\mathcal{C}_u) - m_u; \{u \not\leftrightarrow Y\}] = 0$ by (45).

The next identity is what lets one vertex be removed from T.

Lemma 5.4. (Peeling the Vertex of Largest Rank) *Let T be nonempty, let $k \in T$ have the largest rank, and put $T_0 = T \setminus \{k\}$. For a vertex u set*

$$p(u) = \mathbb{P}\big(\{k \not\leftrightarrow Y \cup T_0\},\ u \leftrightarrow k\big), \qquad \mu(u) = \mathbb{E}\big[F(\mathcal{C}_k)\,;\, \{k \not\leftrightarrow Y \cup T_0\},\ u \leftrightarrow k\big]\,.$$

Then

$$D_u(T) = D_u(T_0) + \mu(u) - p(u)\, m_k\,. \tag{49}$$

Proof. Since k has the largest rank, adjoining k to T_0 leaves each event J_x with $x \in T_0$ unchanged and creates one new event, J_k. On $\{u \leftrightarrow k\}$ the clusters of u and of k coincide, so on that event the conditions defining J_k read $\{k \not\leftrightarrow Y \cup T_0\}$, and $F(\mathcal{C}_u) = F(\mathcal{C}_k)$ there. Comparing (47) for T and for T_0 therefore adds the term $\mu(u)$ to the expectation and subtracts $p(u)m_k$ from the sum. $\square$

5.2 Comparing the quantity $D_u(T)$ at two vertices

The following is where the family of inequalities of Section 4 enters.

Proposition 5.5. (Comparing $D_u(T)$ at Two Vertices) *In the setting above, let $v \notin Y \cup T$. Then for every vertex o,*

$$\mathbb{P}\big(\{v \not\leftrightarrow\ Y \cup T\} \cap \{o \leftrightarrow v\}\big)\, D_v(T) \ \leq\ \mathbb{P}\big(\{v \not\leftrightarrow\ Y \cup T\}\big)\, D_o(T)\,. \tag{50}$$

Proof. Assume first that every hyperedge probability lies strictly between 0 and 1, that $o \notin Y \cup T$ and that $o \neq v$. Let $d_1, \ldots, d_\ell$ be distinct vertices outside $Y \cup T \cup \{o, v\}$, and let M be the functional (29) formed from the set $Y \cup T$, that list, and the two vertices o and v. We prove by induction on the size of T, and simultaneously for every such list and every pair o, v, that

$$M\big[\,u \mapsto D_u(T)\,\big] \geq 0\,. \tag{51}$$

Base case. If T is empty then $D_u(\varnothing) = 0$ for every u by (47), so (51) holds by Lemma 4.8.

Inductive step. Let T be nonempty, let $k \in T$ have the largest rank, put $T_0 = T \setminus \{k\}$, and let p and μ be as in Lemma 5.4. Connection is reflexive, so $p(k) = \mathbb{P}(\{k \nleftrightarrow Y \cup T_0\})$ and $\mu(k) = \mathbb{E}[F(\mathcal{C}_k); \{k \nleftrightarrow Y \cup T_0\}]$. Write

$$\chi(u) = p(k)\, \mu(u) - \mu(k)\, p(u), \qquad \gamma = m_k\, p(k) - \mu(k)\,.$$

Expanding both sides gives the identity

$$p(k)\big(\mu(u) - p(u)\, m_k\big) = \chi(u) - \gamma\, p(u)\,. \tag{52}$$

We claim $M[\chi] \geq 0$ and $M[p] \geq 0$. For the first, let κ be the conditioned covariance (23) formed with the vertex k in place of x, the set $Y \cup T_0$ in place of Y, and the increasing function sending a set of hyperedges to the value of F at the vertex set it spans together with k. Because $(Y \cup T_0) \cup \{k\} = Y \cup T$, the constants (27) and the number (24) attached to this κ are the ones M was formed from. Then $\chi = p(k)^2\, \kappa$, so $M[\chi] = p(k)^2\, M[\kappa]$ by linearity, and $M[\kappa] \geq 0$ is Theorem 4.9. For the second, apply the same theorem with the increasing function that is 1 on nonempty sets of hyperedges and 0 on the empty one. Writing q_0 for the probability that k avoids $Y \cup T_0$ and lies on no open hyperedge, the corresponding covariance is $q_0\, p(u)$ up to the positive factor $p(k)^2$ at every vertex u other than k, which is all that M evaluates, so $q_0\, M[p] \geq 0$. Closing every hyperedge at k has positive probability, so $q_0 > 0$ and $M[p] \geq 0$.

Next we claim

$$\gamma \leq D_k(T_0)\,. \tag{53}$$

By (45) we have $\mathbb{E}[F(\mathcal{C}_k) - m_k; \{k \nleftrightarrow Y\}] = 0$. Splitting $\{k \nleftrightarrow Y\}$ into the part where k avoids T_0 as well and the part where it reaches T_0 turns this into $\gamma = \mathbb{E}[F(\mathcal{C}_k) - m_k; \{k \nleftrightarrow Y\},\ k \leftrightarrow T_0]$. Expanding that restricted expectation over the partition of Lemma 5.3, taken with k in place of u, gives $\gamma = D_k(T_0) + \sum_{x \in T_0} \mathbb{P}(J_x)(m_x - m_k)$, and every summand is at most zero because $r(x) < r(k)$ and r is compatible.

Now apply M to (49), multiply by $p(k) > 0$, and use (52) together with linearity:

$$p(k)\, M\big[\, u \mapsto D_u(T)\, \big] = p(k)\, M\big[\, u \mapsto D_u(T_0)\, \big] + M[\chi] - \gamma\, M[p]\,.$$

The middle term is nonnegative. Since $M[p] \geq 0$, replacing γ by the larger quantity $D_k(T_0)$ of (53) only decreases the right side. Writing $p(u) = p(k)\, c(u)$ with $c(u) = \mathbb{P}(u \in \mathcal{C}_k \mid \{k \nleftrightarrow Y \cup T_0\})$, the resulting expression is $p(k)$ times the value of the functional obtained by inserting k, with the constant c, at the front of the list $d_1, \ldots, d_\ell$, applied to $u \mapsto D_u(T_0)$; this is exactly the recursion (28). That value is nonnegative by the induction hypothesis, applied to the smaller set T_0 with the lengthened list. Dividing by $p(k)$ gives (51).

From (51) to the statement. Take the empty list, so that $M[q] = q(o) - \lambda q(v)$ with $\lambda = \mathbb{P}(o \in \mathcal{C}_v \mid \{v \nleftrightarrow Y \cup T\})$. Then (51) reads $D_o(T) \geq \lambda D_v(T)$, and multiplying by $\mathbb{P}(\{v \nleftrightarrow Y \cup T\}) > 0$ gives (50). If $o = v$ the two sides of (50) are equal. If o lies in Y or in T, the event $\{v \nleftrightarrow Y \cup T\} \cap \{o \leftrightarrow v\}$ is empty, so the left side vanishes while the right side is nonnegative by the two cases recorded after Lemma 5.3.

Parameters in $[0, 1]$. By (48) each $D_u(T)$ is a finite sum over configurations, each summand being a polynomial in the hyperedge probabilities times a value depending on them only through the finitely many numbers m_x; and each m_x is a ratio of polynomials whose denominator is positive at the given parameters. Both sides of (50) are therefore continuous there. Approximate the given parameters by ones lying strictly inside $(0,1)$, choosing for each a compatible rank; by Lemma 5.3 the quantity $D_u(T)$ does not depend on that choice. The case already proved applies to each approximation, and continuity gives (50) in general. $\square$

5.3 Nonnegativity

Theorem 5.6. (Nonnegativity of $D_o(T)$) *In the setting of Section 5.1, $D_o(T) \geq 0$ for every vertex o.*

Proof. We induct on the size of T. If T is empty then $D_o(\varnothing) = 0$.

Let T be nonempty, let $k \in T$ have the largest rank, put $T_0 = T \setminus \{k\}$, and keep p, μ and γ from the proof of Proposition 5.5. Identity (49) with $u = o$ reads $D_o(T) = D_o(T_0) + \mu(o) - m_k\, p(o)$.

Apply the conditional association of Corollary 2.14 with the vertex k, the set $Y \cup T_0$, and the two increasing functions of $\mathcal{K}_k$ given by F evaluated at the spanned vertex set together with k, and by the indicator that o lies in the cluster of k. This gives $p(o)\,\mu(k) \leq p(k)\,\mu(o)$, which rearranges to

$$p(k)\big(m_k\, p(o) - \mu(o)\big) \leq p(o)\big(m_k\, p(k) - \mu(k)\big) = p(o)\,\gamma\,.$$

By (53) and $p(o) \geq 0$ the right side is at most $p(o)\,D_k(T_0)$, and by Proposition 5.5 applied to T_0 with $v = k$ that is at most $p(k)\,D_o(T_0)$. Hence

$$p(k)\,D_o(T) = p(k)\,D_o(T_0) + p(k)\big(\mu(o) - m_k p(o)\big) \geq 0\,.$$

If $p(k) > 0$ this gives $D_o(T) \geq 0$. If $p(k) = 0$ then $p(o) = 0$ and $\mu(o) = 0$, so $D_o(T) = D_o(T_0)$, which is nonnegative by the induction hypothesis. $\square$

5.4 The gluing inequalities

We now prove the inequalities that Section 3 showed to be sufficient. The proof is where the localization at the vertex of least rank pays off, and it delivers the strongest of the three forms directly.

Theorem 5.7. (Multiplicative Gluing Holds) *Let o and b be vertices and let A be a nonempty finite set of vertices. Then*

$$\mathbb{P}(o \leftrightarrow b) \geq \mathbb{P}(o \leftrightarrow A) \cdot \min_{a \in A} \mathbb{P}(a \leftrightarrow b)\,.$$

Proof. We apply Theorem 5.6 to one particular function and then estimate the resulting sum from below. The proof has three steps.

Step 1: choose the data. Take $Y = \varnothing$ and $T = A$ in Section 5.1, and let $F(S) = \mathbf{1}\{b \in S\}$ for a set S of vertices. This function is nonnegative, and it is increasing because enlarging a set can only help it contain b. Its mean at a vertex a is

$$m_a = \mathbb{E}\big[\mathbf{1}\{b \in \mathcal{C}_a\}\big] = \mathbb{P}(a \leftrightarrow b)\,.$$

Choose a compatible rank for A and this F, which exists by the remark after Definition 5.1.

Step 2: apply the nonnegativity. Theorem 5.6 applied with $T = A$ gives $D_o(A) \geq 0$, which by Definition 5.2 reads

$$\sum_{a \in A} \mathbb{P}(J_a)\, m_a \;\leq\; \mathbb{E}\big[F(\mathcal{C}_o)\,;\, o \leftrightarrow A\big]\,. \tag{54}$$

The right side of (54) equals $\mathbb{P}(o \leftrightarrow b \text{ and } o \leftrightarrow A)$, since $F(\mathcal{C}_o)$ is the indicator that o is connected to b.

Step 3: estimate the left side and conclude. Bounding every mean below by the smallest one,

$$\sum_{a \in A} \mathbb{P}(J_a)\, m_a \;\geq\; \Big(\min_{a \in A} m_a\Big) \sum_{a \in A} \mathbb{P}(J_a) \;=\; \Big(\min_{a \in A} \mathbb{P}(a \leftrightarrow b)\Big)\, \mathbb{P}(o \leftrightarrow A)\,,$$

where the last equality is Lemma 5.3. Combining this with (54) and with $\mathbb{P}(o \leftrightarrow b \text{ and } o \leftrightarrow A) \leq \mathbb{P}(o \leftrightarrow b)$ gives the claim. □

Step 3 is where the argument does not degrade, for the following reason. The numbers $\mathbb{P}(J_a)$ sum to $\mathbb{P}(o \leftrightarrow A)$ and not to $|A|$ times anything, because Lemma 5.3 makes them a partition. A union bound over the vertices of A would instead produce $\sum_a \mathbb{P}(a \nleftrightarrow b)$, with one term for each vertex of A and no cancellation. The vertex of least rank is what converts a sum over the vertices of A into a single expectation.

The other two forms follow at once.

Corollary 5.8. (Additive and Near-One Gluing) *Additive gluing, Definition 3.3, and near-one gluing, Definition 3.4, both hold.*

Proof. Theorem 5.7 gives multiplicative gluing, Definition 3.2. Additive gluing follows by Lemma 3.5, and near-one gluing then follows by Lemma 3.6. □

5.5 The consequence for the lattice

Assembling the sections gives the theorem these notes set out to prove.

Theorem 5.9. (The Percolation Probability Vanishes at the Critical Probability) *For every integer $d \geq 2$, Bernoulli bond percolation on $\mathbb{Z}^d$ satisfies $\theta(p_{\mathrm{c}}(d)) = 0$. Equivalently, the percolation probability is continuous on the whole parameter interval.*

Proof. Section 5.2 proved Proposition 5.5 from Theorem 4.9, and Section 5.3 proved Theorem 5.6 from that. Theorem 5.7 and Corollary 5.8 then give all three gluing inequalities, and additive gluing gives the conclusion by Corollary 3.8. The equivalence with continuity is Theorem 1.15. □

Two dimensions was known from Harris [23] and Kesten [24], and dimensions eleven and above from Hara and Slade [21] and Fitzner and van der Hofstad [14]. The argument above covers every dimension at once and in particular the range from three to ten, which no previous method reached.

6 Bond and Site Percolation

Everything proved so far is a statement about a finite family of independent hyperedges. This section identifies bond percolation and site percolation inside that family and reads off what the finite inequality says about each lattice.

The proof of the finite inequality, which is everything in Sections 4 and 5, is due to Anthropic, who gave it for bond percolation. Nothing in it uses the fact that an edge joins exactly two vertices. Running it for hyperedges is ours, together with the reading of every function of a cluster at $\mathcal{K}_S$ that the general model forces, and the site model below.

Section 6.1 gives the two identifications and shows why the one for site percolation needs an extra vertex on each edge, and Section 6.2 draws the two lattice conclusions.

6.1 Bond and site percolation as special cases

Example 6.1 (Bond and Site Percolation) Bond percolation on a graph G is the case of Definition 2.7 in which every incidence set is the pair of endpoints of an edge.

Site percolation on G needs one extra vertex for each edge, and we take G to have no isolated vertex. Take as vertices those of G together with a new vertex m_e for every edge e. Take one hyperedge for each vertex of G, the one at u having incidence set $\{u\} \cup \{\, m_e : e \ni u \,\}$ and being open with the probability that u is open. Since u lies on an edge, that incidence set has at least two elements, as Definition 2.7 requires. Two distinct vertices of G are connected in this model exactly when they are connected in site percolation on G.

Only distinct vertices are covered, and they have to be: connection in Definition 2.7 is reflexive, so u is connected to itself whether or not u is open. For distinct u and v the claim is checked in one step each way. If u and v are neighbors in G and both open, the two hyperedges are open and share m_e, so u and v are connected; concatenating along an open path of G gives the general case. Conversely a chain of open hyperedges from u to v passes through added vertices only, and m_e lies on the hyperedges of the two endpoints of e and on no others, so each such passage crosses an edge of G with both endpoints open.

The extra vertices are needed. Giving the hyperedge of u the incidence set $\{u\}$ together with the neighbors of u does not work: that hyperedge alone would make u adjacent to a neighbor whose own hyperedge is closed, and on a star it would make distinct neighbors of the center adjacent to one another.

Site percolation is not itself a case of bond percolation. Gladkov and Zimin [18] proved that bond percolation cannot simulate site percolation even approximately, already near a vertex of degree three. In the other direction, bond percolation on a graph is site percolation on its line graph, an observation of Fisher and of Fisher and Essam recorded in Kesten [25, §2.5], where the line graph is called the covering graph. So of the two models site percolation is the more general, and a proof written for bond percolation alone does not reach it.

6.2 The consequence for the lattice

Specializing to the two models of Example 6.1 gives the finite inequality for both, from one proof.

Corollary 6.2. (Multiplicative Gluing for Both Models) *Suppose multiplicative gluing holds in the model of Definition 2.7. Then it holds for bond percolation on every finite graph with*

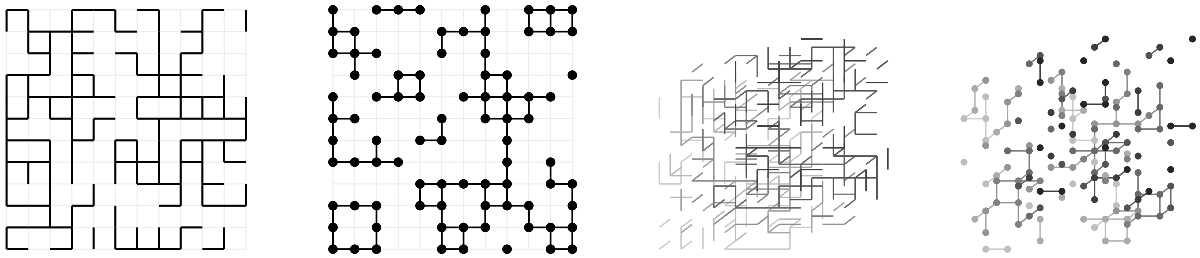


Figure 9: The four models these notes cover, each near its critical parameter. Bond percolation opens edges, site percolation opens vertices and joins two open neighbors. In three dimensions the pictures are drawn in projection, with depth shown by shading. The critical parameter is exactly $1/2$ for bond percolation in two dimensions; the other three values are not known in closed form, and the numbers used here are the usual numerical estimates.

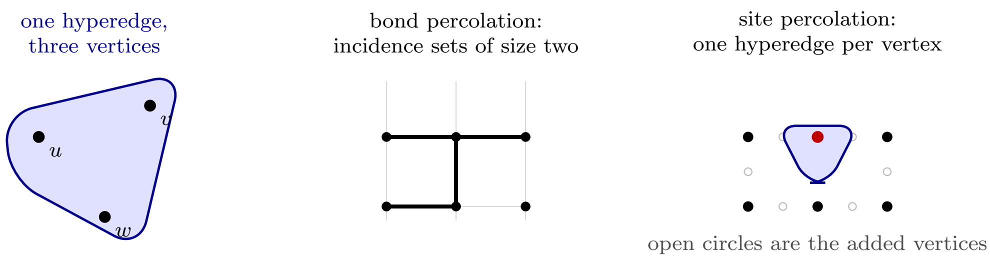


Figure 10: A hyperedge and the two models. On the left, a hyperedge incident to three vertices makes all three mutually adjacent when it is open. In the middle, incidence sets of size two give bond percolation. On the right, the hyperedge of a vertex of G reaches that vertex and the added vertex on each incident edge, so two vertices of G are joined exactly when both hyperedges are open.

independent edge parameters. It also holds for site percolation on every finite graph with no isolated vertex and independent vertex parameters, whenever o and b are distinct and neither of them lies in A.

Proof. Bond percolation is the instance in which every incidence set has two elements, so the first statement is what multiplicative gluing says about that instance.

For the second, take the instance of Example 6.1. The quantities the inequality compares are $\mathbb{P}(o \leftrightarrow b)$, $\mathbb{P}(o \leftrightarrow A)$ and $\mathbb{P}(a \leftrightarrow b)$ for $a \in A$, and under the stated hypotheses each of them compares two distinct vertices of G. By Example 6.1 each is therefore the same number in the two models, so the inequality for the instance is the inequality for site percolation. $\square$

For bond percolation the passage from the finite inequality to the lattice is Theorem 3.7: additive gluing follows from the multiplicative form by Lemma 3.5, and the conclusion is Corollary 3.8. This is Theorem 5.9.

For site percolation the passage is not the bond argument reread. The block construction of Theorem 3.7 has to be carried out again in the site model, and what it delivers is this: if site percolation on $\mathbb{Z}^d$ has an infinite cluster at a parameter p, then site percolation in a slab of finite width has one at the same p. As with Theorem 3.7, we cite this rather than reproduce it, and it is assumed nowhere: both are proved, and checked in Lean, in the development that accompanies these notes, where the final statement carries no hypothesis beyond the dimension.

Granting it, the conclusion follows. In two dimensions the percolation probability of site percolation on $\mathbb{Z}^2$ is continuous on the whole parameter interval, and in particular vanishes at the critical probability, by Russo [30]. For $d \geq 3$, a slab that percolates may be translated into a half-space, which then percolates as well because enlarging the graph cannot destroy an infinite cluster, and the half-space theorem of Barsky, Grimmett, and Newman [5] in its site form says that a half-space does not percolate at the critical parameter. So for every $d \geq 2$ the percolation probability of site percolation on $\mathbb{Z}^d$ vanishes at its critical probability.

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